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Robust out-of-distribution detection under noisy labels

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ABSTRACT

Out-of-distribution (OOD) detection plays a critical role in ensuring the reliability of machine learning systems deployed in real-world applications. While existing OOD detection methods assume clean in-distribution (ID) training data with correct labels, this assumption often fails due to practical issues such as incomplete label taxonomies and human annotation errors. The presence of noisy labels in the training data significantly degrades the ability to distinguish between ID and OOD instances. In this work, we propose a framework that integrates noisy label learning (NLL) with post-hoc OOD detection methods for robust OOD detection under noisy labels. To align NLL methods with OOD detection, we introduce three key propositions: (1) NLL enhances OOD detection under noisy labels; (2) Sharpening model outputs in semi-supervised learning (SSL)-based NLL benefits OOD detection; and (3) Mixup in NLL can degrade OOD detection despite ID accuracy gains. Through extensive experiments on the OpenOOD benchmark across diverse noisy label scenarios, we validate that the proposed framework remarkably improves OOD detection performance under noisy labels.

1. Introduction

Out-of-distribution (OOD) detection refers to the task of identifying instances that fall outside the distribution of the training data. Machine learning models are typically trained under the closed-world assumption, where both training and test instances are drawn from the same distribution, referred to as the in-distribution (ID). Under this assumption, the model is expected to encounter only the classes from ID data during the inference phase. However, in real-world applications, models may be exposed to instances from classes or domains unseen during training, referred to as OOD instances. Such OOD instances can lead to erroneous predictions, thereby compromising the reliability and safety of machine learning systems, particularly in high-stakes domains such as autonomous driving (Filos et al., 2020) and medical diagnosis (Guha Roy et al., 2022; Wang et al., 2017). To address these concerns, research on OOD detection has made significant progress, particularly in image classification tasks (Yang et al., 2024).

Existing OOD detection methods typically assume clean training data with correctly labeled instances (Humblot-Renaux et al., 2024). However, in real-world scenarios, noisy labels often arise due to several practical factors, including incomplete label taxonomies and human annotation errors (Li & Zhu, 2024). A recent study by Humblot-Renaux et al. (2024) demonstrated through extensive benchmarking that the presence of noisy labels in the training data not only degrades the classification accuracy on ID instances but also reduces the separability between ID and OOD instances. This ultimately leads to a substantial performance drop in OOD detection methods, severely limiting their reliability and applicability under noisy label scenarios. Despite its significance, this critical issue has been largely overlooked in prior research.

Fig. 1 shows the OOD detection performance across various noisy label scenarios on CIFAR-10N, CIFAR-100N, and Tiny-ImageNet. When the training dataset contains noisy labels, the OOD detection performance consistently degrades compared to the clean label baseline (*i.e.*, the model trained on clean labels). The degradation becomes more severe as the noise ratio increases. In particular, the performance drop is pronounced in the Near-OOD settings, which are semantically similar to ID instances.

The primary objective of this study is to be among the first to address OOD detection under noisy ID labels, where existing OOD detection methods suffer substantial degradation. We propose a novel framework that leverages noisy label learning (NLL) during the model training phase, which was originally designed to improve classification accuracy in the presence of noisy labels under the closed-world assumption. The trained model is then coupled with post-hoc OOD detection methods to enable accurate and reliable detection of OOD instances. Beyond a straightforward combination of NLL and OOD detection, we identify two key factors arising from the inherent incompatibilities between NLL methods and OOD detection: the beneficial effect of model output sharpening for clean instances and the detrimental effect of Mixup-related components. Building upon these insights, we derive simple yet effective adaptations tailored to this setting to further enhance ID-OOD separa-

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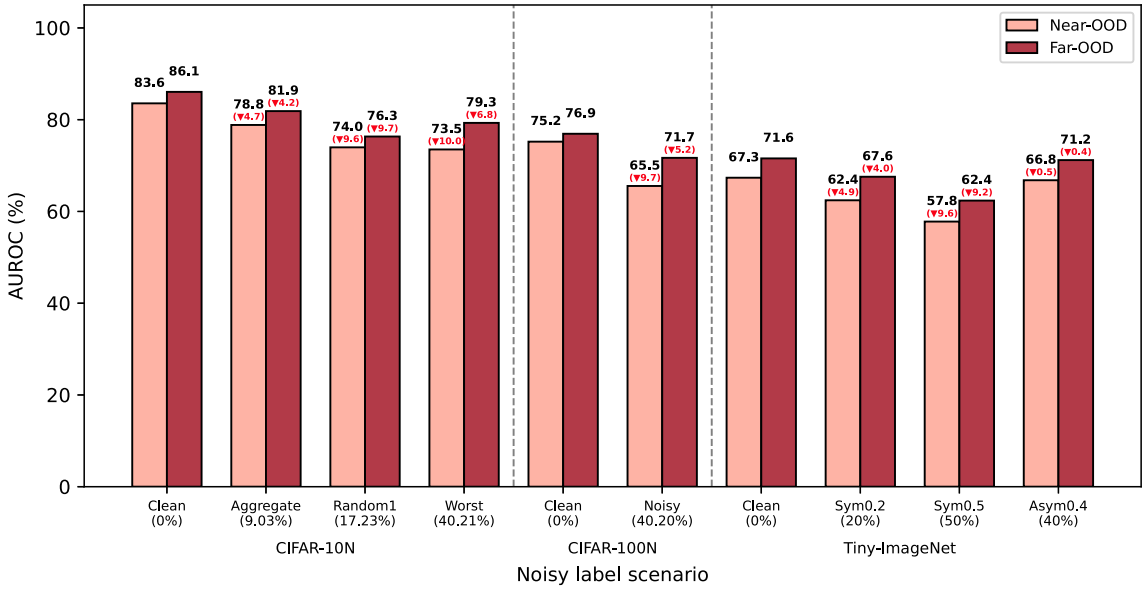


Fig. 1. Degradation of OOD detection performance under various noisy label scenarios on CIFAR-10N, CIFAR-100N, and Tiny-ImageNet. Models are trained using standard cross-entropy loss on the respective training datasets. OOD detection performance is evaluated in terms of AUROC in both Near-OOD and Far-OOD settings, following the evaluation protocol described in Section 5.

bility across a broad range of NLL methods and post-hoc OOD detectors under various noisy label scenarios.

For the proposed framework, we present three key propositions: (1) NLL enhances OOD detection under noisy labels; (2) Sharpening model outputs in semi-supervised learning (SSL)-based NLL benefits OOD detection; and (3) Mixup in NLL can degrade OOD detection despite ID accuracy gains. We empirically validate these propositions through comprehensive experiments on the OpenOOD benchmark (Zhang et al., 2024). Our findings corresponding to the three propositions are summarized as follows:

- Incorporating NLL not only improves ID classification but also enhances OOD detection, even under severe noisy label scenarios.
- Sharpening model outputs via energy minimization on instances predicted to have clean labels in SSL-based NLL methods improves OOD detection.
- Removing Mixup-related components from NLL methods improves OOD detection, albeit at the cost of ID classification accuracy.

The remainder of this paper is organized as follows. In Section 2, we review related work on OOD detection and NLL. In Section 3, we describe the NLL methods investigated in this study. In Section 4, we present the proposed framework along with our propositions. In Section 5, we detail the benchmark datasets and the evaluation protocol. In Section 6, we provide experimental results and discussions. Finally, conclusions and future work are presented in Section 7.

2. Related work

2.1. Out-of-distribution (OOD) detection

OOD detection is formulated as the task of determining whether a query instance $\mathbf{x}_*$ is OOD based on an OOD score $s(\mathbf{x}_*; f_\theta)$ computed by a model f_θ. The training dataset $\mathcal{D}_{\text{trn}} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^N$ is assumed to consist only of ID instances, where $\mathbf{x}_i$ denotes an image and $\mathbf{y}_i \in \{0,1\}^C$ is a one-hot encoded class label for one of the predefined classes C. As a precondition, the model f_θ is a neural network whose output $f_\theta(\mathbf{x}_*) \in [0,1]^C$ represents the softmax probabilities of the C classes. The corresponding logit vector $\mathbf{z}_*$ is the intermediate output before softmax activation. The OOD score $s(\mathbf{x}_*; f_\theta)$ quantifies the degree of deviation from ID represented by $\mathcal{D}_{\text{trn}}$. The goal of an OOD detection method is to maintain high classification accuracy on ID instances while reliably distinguishing OOD instances (Zhang et al., 2024). If $s(\mathbf{x}_*; f_\theta) > \tau$ for a certain threshold τ, the instance $\mathbf{x}_*$ is identified as OOD, indicating that it does not belong to any of the predefined ID classes.

Existing OOD detection methods can be categorized into three groups (Zhang et al., 2024). *Post-hoc methods* are applied after a model has been trained without requiring any additional modification or retraining. OOD scores are computed using model outputs, such as softmax probabilities, logit vectors, feature representations, or gradients, for a query instance $\mathbf{x}_*$. This plug-and-play nature makes post-hoc methods highly practical and broadly compatible with any classifier. *Training-time regularization methods* directly alter the training process through architectural modifications or auxiliary regularization terms in the learning objective to improve OOD detection capability. These methods operate under the assumption that OOD data are not available during training. *Outlier exposure methods* train a model by extending the training dataset with external or synthetic OOD instances that mimic potential future OOD instances to learn the distinction between ID and OOD instances during training.

2.2. Taxonomy of post-Hoc OOD detection methods

We adopt post-hoc OOD detection methods to ensure compatibility with any models trained using NLL. Post-hoc methods are further categorized according to how their OOD scoring functions are designed and computed (Humblot-Renaux et al., 2024).

- **Softmax-based OOD detection** relies on the assumption that ID instances yield higher softmax confidence and lower entropy than OOD instances, as neural networks tend to yield uncertain predictions for instances that were not encountered during training. A query instance is identified as OOD if it exhibits low softmax confidence or high entropy. Methods in this category include MSP (Hendrycks & Gimpel, 2017), TempScale (Guo et al., 2017), and ODIN (Liang et al., 2018).
- **Logit-based OOD detection** is based on an idea similar to softmax-based OOD detection but operates on logit vectors to avoid information loss. Methods in this category include MLS (Hendrycks et al., 2022), EBO (Liu et al., 2020b), ReAct (Sun et al., 2021), Rank-

Feat (Song et al., 2022b), DICE (Sun & Li, 2022), and ASH (Djurisic et al., 2023).

- **Distance-based OOD detection** measures the extent to which a query instance deviates from ID instances in the model's intermediate feature space, where OOD instances are expected to lie far from the feature manifold formed by ID instances. A query instance is identified as OOD if it exhibits a large distance from the training dataset in the feature space. Methods in this category include MDS (Lee et al., 2018), MDSEns (Lee et al., 2018), RMDS (Ren et al., 2021), KLM (Hendrycks et al., 2022), OpenMax (Bendale & Boult, 2016), SHE (Zhang et al., 2023a), Gram (Sastry & Oore, 2020), KNN (Sun et al., 2022), and VIM (Wang et al., 2022).
- **Gradient-based OOD detection** computes OOD scores based on the gradients of the model, assuming that ID and OOD instances induce distinguishable gradient patterns. A query instance is identified as OOD if it exhibits gradient patterns different from those of the training data (*e.g.*, smaller gradient norm). Methods in this category include GradNorm (Huang et al., 2021).

2.3. Noisy label learning (NLL)

NLL aims to improve the classification accuracy of a model when the training dataset D_{trn} contains noisy labels, where some instances are incorrectly labeled. Under such noisy conditions, neural networks tend to memorize label noise, which degrades their generalization performance (Zhang et al., 2021). NLL addresses this issue by making the training process robust to noisy labels. Existing NLL methods can be categorized into four approaches based on how they handle noisy labels during training (Li & Zhu, 2024).

- **Noise transition matrix estimation approach** models the label corruption process using a noise transition matrix, where each element indicates the probability of one class being mislabeled as another. Some methods add learnable layers (Goldberger & Ben-Reuven, 2017; Sukhbaatar et al., 2015), while others estimate a noise transition matrix based on model predictions or small clean subsets (Hendrycks et al., 2018; Patrini et al., 2017). However, these methods typically assume uniform corruption across all instances, which limits their applicability in real-world scenarios.
- **Sample selection approach** identifies and filters out potentially mislabeled instances from the training dataset and trains the model on the remaining clean instances. Decoupling (Malach & Shalev-Shwartz, 2017) selects clean instances based on prediction disagreement between two neural networks, while others utilize small-loss training instances as clean instances (Han et al., 2018; Huang et al., 2019; Jiang et al., 2018; Shen & Sanghavi, 2019).
- **Semi-supervised learning (SSL)-based approach** leverages SSL by treating instances with noisy labels as an unlabeled set and clean instances as a labeled set to maximize data utility. The key to this approach lies in accurately distinguishing between clean and noisy instances, as the model is trained on both sets using techniques such as pseudo-labeling or label correction applied to the unlabeled set. Examples include DivideMix (Li et al., 2020), ProMix (Xiao et al., 2023), PGDF (Chen et al., 2023), and TCL (Huang et al., 2023).
- **Noise-robust regularization approach** introduces regularization techniques to mitigate overfitting to noisy labels, encouraging the model to generalize rather than memorize noisy labels. Examples include REGCL (Li & Zhang, 2021), PHuber (Menon et al., 2020), ELR (Liu et al., 2020a), and SOP (Liu et al., 2022).

In this study, we focus on the noise-robust regularization and SSL-based approaches, both of which have demonstrated state-of-the-art performance on NLL benchmarks. These approaches are used to train neural networks that are subsequently paired with post-hoc OOD detection methods in our proposed framework. Representative methods for each approach are introduced in the following section.

3. Preliminary

We investigate the effectiveness of the proposed framework using five NLL methods. In this study, we select top-performing methods from two approaches: SOP (Liu et al., 2022) and ELR (Liu et al., 2020a) from the noise-robust regularization approach, and ProMix (Xiao et al., 2023), PGDF (Chen et al., 2023), and TCL (Huang et al., 2023) from the SSL-based approach. Fig. 2 provides a schematic overview of these methods. We briefly describe the training architectures below.

3.1. Sparse over-parameterization (SOP)

SOP (Liu et al., 2022) explicitly models instance-wise label noise to mitigate its negative impact on training. It introduces a learnable noise estimate $\mathbf{s}_i$ for each instance $(\mathbf{x}_i, \mathbf{y}_i)$ to adjust the output of the model f_θ as:

$$\mathbf{y}_i \approx \phi(\mathbf{z}_i + \mathbf{s}_i), \quad \text{where} \quad \mathbf{s}_i = \mathbf{u}_i \odot \mathbf{u}_i \odot \mathbf{y}_i - \mathbf{v}_i \odot \mathbf{v}_i \odot (\mathbf{1} - \mathbf{y}_i), \tag{1}$$

where $\mathbf{z}_i$ denotes the logit vector, ϕ is a softmax-like normalization function, and $\odot$ denotes the Hadamard product. The noise correction term $\mathbf{s}_i$ for each instance $(\mathbf{x}_i, \mathbf{y}_i)$ is parameterized by instance-wise parameters $\mathbf{u}_i$ and $\mathbf{v}_i$, which selectively activate for noisy labels and promote sparsity. During training, the model parameters θ and the noise-enhancing parameters $\mathbf{u}_i$ are updated via cross-entropy loss $\mathcal{L}_{\text{CE}}$, while the noise-suppressing parameters $\mathbf{v}_i$ are updated using mean squared error loss $\mathcal{L}_{\text{MSE}}$.

Fig. 2 (a) provides an overview of the SOP training architecture. In this study, we adopt SOP+, an improved variant of SOP that incorporates consistency and class-balance regularization to further enhance robustness. The source code of SOP is publicly available at https://github.com/shengliu66/SOP.

3.2. Early-learning regularization (ELR)

ELR (Liu et al., 2020a) exploits the early-learning phenomenon, where a model initially learns from clean labels but gradually memorizes noisy labels as training progresses. To mitigate this memorization, ELR incorporates a regularization term that preserves the knowledge acquired during the early training stage. Specifically, it uses a target probability vector $\mathbf{t}_i$ for each instance $(\mathbf{x}_i, \mathbf{y}_i)$, which represents a smoothed history of the model predictions from previous steps. The target vector $\mathbf{t}_i^{(k)}$ at the k-th epoch is updated via an exponential moving average (EMA) to retain the clean learning signal from early predictions:

$$\mathbf{t}_i^{(k)} = \beta\, \mathbf{t}_i^{(k-1)} + (1-\beta)\, f_\theta^{(k)}(\mathbf{x}_i), \tag{2}$$

where $0 \le \beta < 1$ is a momentum hyperparameter and $f_\theta^{(k)}(\mathbf{x}_i)$ denotes the predicted softmax probabilities at epoch k. The overall training objective combines the cross-entropy loss $\mathcal{L}_{\text{CE}}$ with a regularization term:

$$\mathcal{L}_{\text{ELR}} = \mathcal{L}_{\text{CE}} + \frac{\lambda_{\text{ELR}}}{N} \sum_{i=1}^{N} \log\left(1 - f_\theta(\mathbf{x}_i) \cdot \mathbf{t}_i\right), \tag{3}$$

where λ_{ELR} controls the regularization strength. This regularization term guides the model f_θ to remain aligned with its early-stage predictions, which are less influenced by label noise.

Fig. 2 (b) provides an overview of the ELR training architecture. In this study, we adopt the original ELR, although there is an improved variant ELR+, which integrates weight averaging, two-network training, and Mixup. The source code of ELR is publicly available at https://github.com/shengliu66/ELR.

3.3. ProMix

ProMix (Xiao et al., 2023) alternates between *progressive instance selection* and *debiased SSL* to effectively identify clean instances and mitigate biases during training.

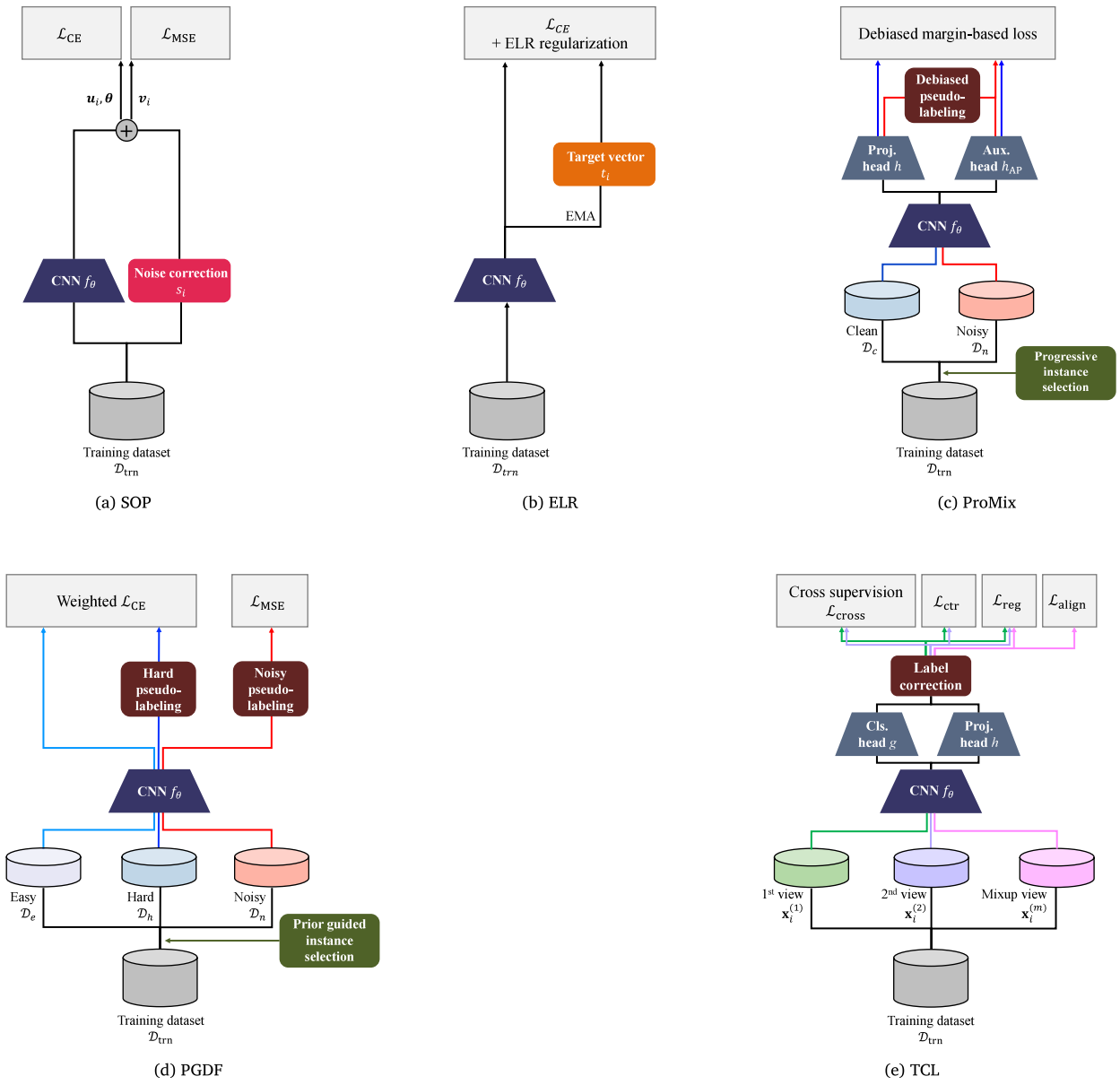


Fig. 2. Overview of NLL methods investigated in this study: (a) SOP; (b) ELR; (c) ProMix; (d) PGDF; and (e) TCL.

In the *progressive instance selection* step, the training dataset $\mathcal{D}_{\text{trn}}$ is progressively separated into a clean subset $\mathcal{D}_{\text{c}}$ and a noisy subset $\mathcal{D}_{\text{n}}$. This separation is based on the observation that clean instances typically have lower training loss and higher softmax confidence. The clean subset $\mathcal{D}_{\text{c}}$ serves as the labeled subset and the noisy subset $\mathcal{D}_{\text{n}}$ as the unlabeled subset for subsequent training.

In the *debiased SSL* step, the model f_θ is trained on both $\mathcal{D}_{\text{c}}$ and pseudo-labeled $\mathcal{D}_{\text{n}}$, while addressing two types of bias: confirmation bias from erroneous pseudo-labels and distribution bias from class imbalance in $\mathcal{D}_{\text{c}}$. To mitigate confirmation bias, an auxiliary head h_{AP} decouples instance selection from training, which prevents reinforcement of selection error. Specifically, the primary projection head h is trained only on $\mathcal{D}_{\text{c}}$, while the auxiliary projection head h_{AP} is trained on both $\mathcal{D}_{\text{c}}$ and pseudo-labeled $\mathcal{D}_{\text{n}}$ by debiased pseudo-labeling. To address distribution bias, both h and h_{AP} employ a debiased margin-based loss that adjusts the logits using dynamically estimated class priors. This encourages the model to focus less on over-represented classes and more on under-represented ones. The training objective combines losses from both heads and incorporates consistency regularization and Mixup applied to $\mathcal{D}_{\text{c}}$.

Fig. 2 (c) provides an overview of the ProMix training architecture. The source code of ProMix is publicly available at https://github.com/Justherozen/ProMix.

3.4. Prior-guided denoising framework (PGDF)

PGDF (Chen et al., 2023) alternates between *prior-guided instance selection* and *denoising SSL* to identify clean instances and apply subset-specific pseudo-labeling.

In the *prior-guided instance selection* step, the training dataset $\mathcal{D}_{\text{trn}}$ is divided into three subsets based on prediction histories: an easy subset $\mathcal{D}_{\text{e}}$, a hard subset $\mathcal{D}_{\text{h}}$, and a noisy subset $\mathcal{D}_{\text{n}}$. The primary model f_θ is first trained to obtain prediction histories, which are used to partition instances into an easy subset $\mathcal{D}_{\text{e}}$, a noisy subset $\mathcal{D}_{\text{n}}$, or an ambiguous subset for the next step. A dedicated classifier then assigns these ambiguous instances to a hard subset $\mathcal{D}_{\text{h}}$ or a noisy subset $\mathcal{D}_{\text{n}}$.

In the *denoising SSL* step, a label transition matrix estimated from $\mathcal{D}_e$ is used to calibrate and refine pseudo-labels. Subset-specific pseudo-labeling strategies are then applied: instances in $\mathcal{D}_e$ retain their original labels, instances in $\mathcal{D}_n$ are assigned refined pseudo-labels (noisy pseudo-labeling), and instances in $\mathcal{D}_h$ are assigned a weighted combination of their observed labels and pseudo-labels (hard pseudo-labeling) to balance prediction confidence and label reliability. The model f_θ is trained using MixMatch (Berthelot et al., 2019b), which incorporates Mixup. Here, $\mathcal{D}_e$ and $\mathcal{D}_h$ are treated as labeled subsets, while $\mathcal{D}_n$ is treated as an unlabeled subset, leading to a different loss function. The overall training objective consists of a weighted cross-entropy loss $\mathcal{L}_{CE}$ computed on $\mathcal{D}_e$ and $\mathcal{D}_h$, a mean squared error loss $\mathcal{L}_{MSE}$ on $\mathcal{D}_n$, and a regularization term applied across all subsets.

Fig. 2 (d) provides an overview of the PGDF training architecture. The source code of PGDF is publicly available at https://github.com/bupt-ai-cz/PGDF.

3.5. Twin contrastive learning (TCL)

TCL (Huang et al., 2023) alternates between *label correction* and *robust representation learning* to estimate clean probabilities and learn representations that are robust to label noise.

In the *label correction* step, feature representations for all instances in the training dataset $\mathcal{D}_{trn}$ are extracted using the model f_θ and the representation head h. A class-wise Gaussian mixture model (GMM) is fitted in the representation space, with its cluster assignments guided by the predictions of the classification head g, to estimate the probability that each instance belongs to its annotated class. A two-component GMM is then fitted to the agreement scores between the inferred cluster assignments and the class labels, yielding a clean probability w_i for each instance $(\mathbf{x}_i, \mathbf{y}_i) \in \mathcal{D}_{trn}$. Instances with high w_i values can be regarded as clean instances. The prediction target for each instance is constructed by interpolating the annotated class label and the model prediction with weights w_i and $1 - w_i$, respectively, thereby reducing the influence of potentially corrupted labels.

In the *robust representation learning* step, the model f_θ and its heads g and h are trained using the following loss terms. The cross-supervision loss $\mathcal{L}_{cross}$ computes the classification loss twice by swapping the prediction targets for two randomly augmented views $\mathbf{x}_i^{(1)}$ and $\mathbf{x}_i^{(2)}$ of each instance $(\mathbf{x}_i, \mathbf{y}_i)$, encouraging the prediction for each view to match the prediction target generated from the other view. The entropy regularization term $\mathcal{L}_{reg}$ is employed to prevent degenerate solutions and encourage confident predictions. The InfoNCE loss $\mathcal{L}_{ctr}$ is applied to the representations of the two augmented views for contrastive learning. The alignment loss $\mathcal{L}_{align}$ is computed using Mixup augmented instances to improve within-class compactness and enhance class separability in the representation space. Finally, the overall training objective is given by

$$\mathcal{L}_{TCL} = \mathcal{L}_{cross} + \mathcal{L}_{reg} + \mathcal{L}_{ctr} + \mathcal{L}_{align}. \tag{4}$$

Fig. 2 (e) provides an overview of the TCL training architecture. The source code of TCL is publicly available at https://github.com/Hzzone/TCL.

4. Proposed framework

4.1. Overview

We consider a scenario where the training dataset $\mathcal{D}_{trn}$ contains only ID instances, but some of their labels are corrupted by label noise. The primary objective is to accurately identify whether a query instance $\mathbf{x}_*$ is OOD, even in the presence of noisy labels in the training dataset $\mathcal{D}_{trn}$.

Fig. 3 compares the ID–OOD separability of models trained under clean and noisy label scenarios without any NLL methods, evaluated using softmax confidence score distribution and t-SNE visualization of feature representations. The model trained on a clean dataset, characterized by high separability among ID classes, shows clear separation between ID and OOD instances. In contrast, the model trained on a noisy dataset, characterized by reduced separability among ID classes, shows significant overlap between ID and OOD instances. The overlap becomes more pronounced as the noise ratio increases. This implies that the presence of noisy labels in the training data impairs the ability of OOD detection methods to reliably assess the OODness of a new query instance.

In our proposed framework, we integrate NLL methods, which are originally designed to improve classification accuracy on ID instances, into the OOD detection pipeline to improve both ID classification and OOD detection performance. A model f_θ is first trained on $\mathcal{D}_{trn}$ using an NLL method to improve robustness to noisy labels. Once trained, the model f_θ is coupled with a post-hoc OOD detection method that assigns an OOD score $s(\mathbf{x}_*; f_\theta)$ to a query instance $\mathbf{x}_*$. The final prediction $\hat{y}_*$ is given by:

$$\hat{y}_* = \begin{cases} \text{OOD}, & \text{if } s(\mathbf{x}_*; f_\theta) > \tau \\ \underset{c \in \{1,\dots,C\}}{\arg\max}\, f_\theta(\mathbf{x}_*)_c & \text{otherwise} \end{cases}, \tag{5}$$

where τ is the decision threshold that distinguishes ID and OOD instances.

4.2. Propositions for the proposed framework

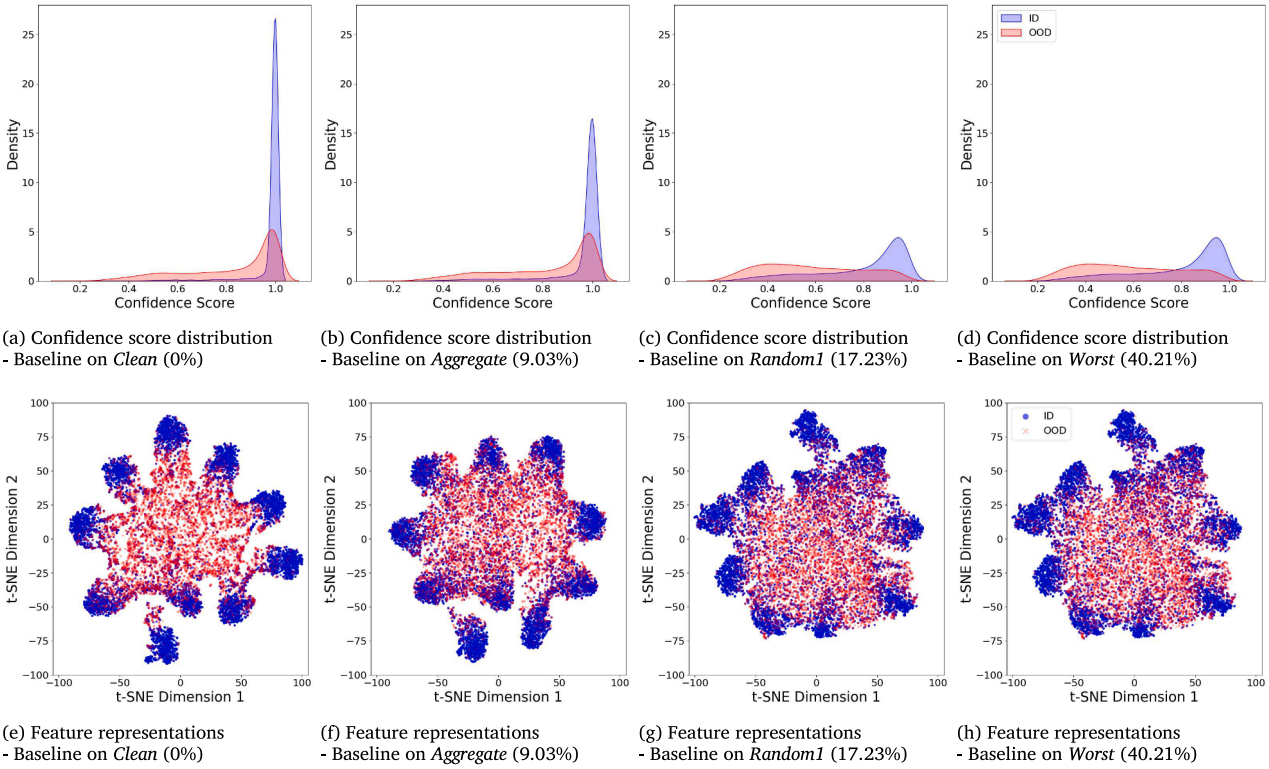
For the proposed framework, we present three key propositions for strengthening OOD detection under noisy label scenarios. Proposition 1 is primarily related to the validity of the proposed framework. Since existing NLL methods were not originally designed for OOD detection, Propositions 2 and 3 introduce adaptation strategies to better align them with OOD detection.

Proposition 1 (NLL Enhances OOD Detection Under Noisy Labels). The presence of noisy labels in the training dataset $\mathcal{D}_{trn}$ degrades ID classification accuracy by blurring the boundaries between ID classes (Li & Zhu, 2024; Song et al., 2022a). An empirical study shows that noisy labels also hinder the ability to separate OOD from ID (Humblot-Renaux et al., 2024). As OOD detection relies on well-disentangled feature representations and calibrated confidence estimates, noisy labels compromise these capabilities and reduce OOD detection performance.

We hypothesize that NLL not only improves ID classification accuracy but also facilitates more reliable OOD detection. NLL mitigates the adverse effects of noisy labels during training, encouraging the model to form more distinct boundaries between ID classes and reducing the number of misclassified ID instances (Li & Zhu, 2024; Song et al., 2022a). These improvements in feature disentanglement and confidence calibration are expected to translate into more robust OOD detection.

Despite this expected benefit, existing NLL methods are not inherently designed or optimized for OOD detection. To overcome this limitation, we propose two task-specific adaptation strategies that refine NLL methods for OOD detection, as detailed in Propositions 2 and 3.

Proposition 2 (Sharpening Model Outputs in SSL-Based NLL Benefits OOD Detection). Sharpening model outputs through entropy or energy minimization is a common strategy in SSL to improve classification accuracy on ID instances (Kim, 2021). This approach facilitates more reliable pseudo-labeling on unlabeled data by promoting confident predictions in SSL (Berthelot et al., 2019a,b; Sohn et al., 2020). Energy minimization has also been shown to directly improve OOD detection by increasing the separation between ID and OOD instances (Liu et al., 2020b). Specifically, applying it to ID instances during training shapes the model's predictive distribution by encouraging them to have consistently lower energy scores. Since OOD instances are not directly constrained by this objective, it can potentially widen the energy gap between ID and OOD instances.



(a) Confidence score distribution - Baseline on *Clean* (0%)
(b) Confidence score distribution - Baseline on *Aggregate* (9.03%)
(c) Confidence score distribution - Baseline on *Random1* (17.23%)
(d) Confidence score distribution - Baseline on *Worst* (40.21%)
(e) Feature representations - Baseline on *Clean* (0%)
(f) Feature representations - Baseline on *Aggregate* (9.03%)
(g) Feature representations - Baseline on *Random1* (17.23%)
(h) Feature representations - Baseline on *Worst* (40.21%)

Fig. 3. Impact of noisy labels on ID-OOD separability. Models are trained using standard cross-entropy loss under different noisy label scenarios on CIFAR-10N. For visualization, ID instances are drawn from CIFAR-10, and OOD instances from CIFAR-100. (a-d) show the softmax confidence distributions. (e-h) show the t-SNE visualizations of feature representations extracted from the penultimate layer.

Motivated by this, we adopt model output sharpening via energy minimization within SSL-based NLL methods, such as ProMix, PGDF, and TCL, for robust OOD detection. Since energy minimization tends to sharpen the model's predictive distribution, applying it to instances with noisy labels may undesirably strengthen the model's confidence in incorrect labels. Therefore, instead of applying energy minimization to the entire training dataset, we apply it exclusively to the clean subset of instances, which we denote as the labeled set $\mathcal{D}_{\mathrm{L}}$. Specifically, we set $\mathcal{D}_{\mathrm{L}} = \mathcal{D}_{\mathrm{c}}$ for ProMix, $\mathcal{D}_{\mathrm{L}} = \mathcal{D}_{\mathrm{e}} \cup \mathcal{D}_{\mathrm{h}}$ for PGDF, and $\mathcal{D}_{\mathrm{L}} = \{(\mathbf{x}_i, \mathbf{y}_i) \in \mathcal{D}_{\mathrm{trn}} | w_i > 0.7\}$ for TCL. We incorporate this sharpening term into the training objective of SSL-based NLL methods. With this addition, the modified training objective becomes:

$$\tilde{\mathcal{L}}_{\mathrm{NLL}} = \mathcal{L}_{\mathrm{NLL}} + \frac{\gamma}{|\mathcal{D}_{\mathrm{L}}|} \sum_{(\mathbf{x}_i, \mathbf{y}_i) \in \mathcal{D}_{\mathrm{L}}} \left[-\log \mathrm{sum}(\exp(\mathbf{z}_i)) \right], \tag{6}$$

where $\mathcal{L}_{\mathrm{NLL}}$ is the original training objective and γ is a hyperparameter that controls the sharpening strength.

Proposition 3 (Mixup in NLL Can Degrade OOD Detection Despite ID Accuracy Gains). Mixup generates synthetic training instances $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ by linearly interpolating a pair of randomly selected instances $(\mathbf{x}_i, \mathbf{y}_i)$ and $(\mathbf{x}_j, \mathbf{y}_j) \in \mathcal{D}_{\mathrm{trn}}$ as follows:

$$\tilde{\mathbf{x}} = \lambda \mathbf{x}_i + (1-\lambda)\mathbf{x}_j; \quad \tilde{\mathbf{y}} = \lambda \mathbf{y}_i + (1-\lambda)\mathbf{y}_j, \tag{7}$$

where the mixing coefficient λ is sampled from the Beta distribution $\mathrm{Beta}(\alpha, \alpha)$ with the shaping hyperparameter α. Training a model with Mixup augmented instances has been shown to improve ID classification accuracy by regularizing the model through interpolated instances between different classes (Zhang et al., 2018).

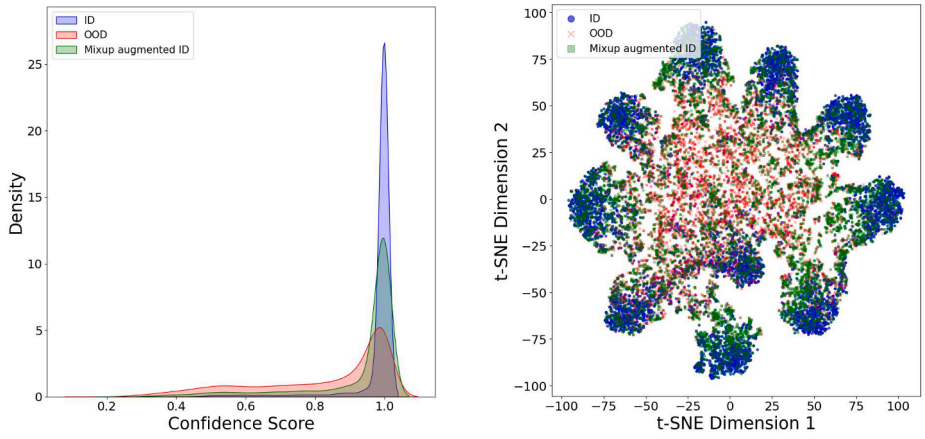
However, the effect of Mixup on OOD detection tends to be detrimental, as it introduces ambiguity between ID and OOD instances. The synthetic instances produced using Mixup, which lie between two classes in both input and label spaces, can overlap with real ID instances (Guo et al., 2019). Additionally, Mixup encourages feature representations to exhibit greater variance, which inadvertently blurs the distinction between ID and OOD instances (Ravikumar et al., 2023). As shown in Fig. 4, Mixup-augmented ID instances are distributed between ID and OOD instances with broad variation, blurring the boundary between ID and OOD in both the softmax output space and feature space. The ambiguous nature of Mixup-augmented ID instances has also been exploited in recent studies to synthesize OOD instances (Yang et al., 2023; Zhang et al., 2023b). The ability of these synthetic instances to serve as OOD proxies reinforces the concern that Mixup in NLL may reduce the model's capacity for reliable OOD detection.

Mixup also leads to underconfident predictions due to the use of smoothed target labels, which are known to compromise OOD detection (Pinto et al., 2022). The cross-entropy between a target distribution $\mathbf{y}$ and a prediction distribution $\hat{\mathbf{y}}$ can be decomposed as:

$$\mathrm{H}(\mathbf{y}, \hat{\mathbf{y}}) = \mathrm{H}(\mathbf{y}) + D_{\mathrm{KL}}(\mathbf{y} || \hat{\mathbf{y}}). \tag{8}$$

By Gibbs' inequality, this loss is minimized when $\hat{\mathbf{y}} = \mathbf{y}$. For a hard-labeled instance $(\mathbf{x}, \mathbf{y})$, where $\mathbf{y}$ represents a one-hot target distribution, minimizing the loss encourages the model's confidence score (*i.e.*, the maximum element of $\hat{\mathbf{y}}$) to approach 1. In contrast, for a Mixup augmented instance $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ obtained using Eq. (7), where $\tilde{\mathbf{y}}$ represents a soft target distribution, the maximum achievable confidence score is limited to $\max(\lambda, 1-\lambda) < 1$. Mixup therefore imposes a ceiling on confidence scores, causing the model to produce less confident predictions on ID instances and thereby diminishing their separability from OOD instances.

Based on these insights, we hypothesize that removing Mixup-related components from NLL methods, such as ELR, ProMix, PGDF, and TCL, can substantially enhance OOD detection performance without significantly compromising ID classification accuracy. For ELR, we use its original version instead of the improved variant ELR+ that incorporates Mixup. For ProMix, we remove the regularization terms based on Mixup and FMix (Harris et al., 2020) from the training objective. For



(a) Confidence score distribution
- Baseline on *Clean* (0%)

(b) Feature representations
- Baseline on *Clean* (0%)

Fig. 4. Relationship between ID, OOD, and Mixup augmented ID instances. The model is trained using standard cross-entropy loss on CIFAR-10 under a clean label scenario. For visualization, ID and Mixup augmented ID instances are drawn from CIFAR-10, while OOD instances are from CIFAR-100. (a) shows the softmax confidence distributions. (b) shows the t-SNE visualizations of feature representations extracted from the penultimate layer.

PGDF, we do not apply Mixup to the subset of instances treated as labeled data (*i.e.*, $\mathcal{D}_e \cup \mathcal{D}_h$) during the MixMatch procedure. For TCL, we replace Mixup augmented instances with weakly augmented instances when computing the training objective.

5. Empirical investigation

5.1. Benchmark datasets

We empirically investigated the three propositions within the proposed framework through experiments on the OpenOOD benchmark (Zhang et al., 2024). We used three datasets with different noisy label scenarios as ID datasets: CIFAR-10N, CIFAR-100N, and Tiny-ImageNet. These scenarios are characterized by different noise mechanisms and varying noise ratios, simulating diverse real-world situations. Each ID dataset was paired with several Near-OOD and Far-OOD datasets. The datasets in the Near-OOD group are semantically similar to the ID dataset but consist of different classes, making them harder to distinguish. In contrast, the datasets in the Far-OOD group deviate significantly in both semantic and statistical properties, making them easier to detect than Near-OOD. To avoid semantic overlap with the ID datasets, certain instances were excluded from the OOD datasets. The details of the ID datasets used, their noise types, and the corresponding OOD datasets are described below.

- **CIFAR-10N** is a noisy version of the CIFAR-10 dataset that incorporates human-annotated noisy labels (Wei et al., 2022). This dataset consists of 32×32 images across 10 balanced object classes with 5000 training images and 1000 test images per class. It provides four types of noisy label scenarios that reflect increasing degrees of label corruption: *Clean* (0%), *Aggregate* (9.03%), *Random1* (17.23%), and *Worst* (40.21%). For OOD detection, the Near-OOD group includes CIFAR-100 and Tiny-ImageNet, and the Far-OOD group includes MNIST, SVHN, Textures, and Places365.
- **CIFAR-100N** extends the CIFAR-100 dataset by incorporating human-annotated noisy labels, similar to CIFAR-10N (Wei et al., 2022). This dataset consists of 32×32 images across 100 balanced object classes with 500 training images and 100 test images per class. It provides two types of noisy label scenarios that reflect increasing degrees of label corruption: *Clean* (0%) and *Noisy* (40.20%). For OOD detection, the Near-OOD group includes CIFAR-10 and Tiny-ImageNet, and the Far-OOD group is the same as that used for CIFAR-10N.
- **Tiny-ImageNet** is a subset of ImageNet-1K and consists of 64×64 resized images across 200 balanced classes with 500 training images and 50 validation images per class. We use the original validation set as the test set. As the dataset provides only clean labels, we synthetically generate symmetric (*Sym*) and asymmetric (*Asym*) label noise following the noise generation protocol in Tanaka et al. (2018). It provides four types of noisy label scenarios: *Clean* (0%), *Sym0.2* (20%), *Sym0.5* (50%), and *Asym0.4* (40%). As Tiny-ImageNet is not included as an ID dataset in the OpenOOD benchmark, the setting used for ImageNet-1K was adopted for OOD detection, where the Near-OOD group includes SSB-Hard and NINCO, and the Far-OOD group includes iNaturalist, Textures, and OpenImage-O.

5.2. Evaluation protocol

Both ID and OOD datasets were split following the OpenOOD benchmark (Zhang et al., 2024). Each ID dataset was partitioned into a training dataset $\mathcal{D}^{\text{ID}}_{\text{trn}}$, a validation set $\mathcal{D}^{\text{ID}}_{\text{val}}$, and a test set $\mathcal{D}^{\text{ID}}_{\text{tst}}$. The training split was used as $\mathcal{D}^{\text{ID}}_{\text{trn}}$. The test split was further divided into 10% for $\mathcal{D}^{\text{ID}}_{\text{val}}$ and 90% for $\mathcal{D}^{\text{ID}}_{\text{tst}}$. The validation set $\mathcal{D}^{\text{ID}}_{\text{val}}$ was used for hyperparameter tuning of OOD detection methods. For each OOD dataset, a subset was set aside as a validation set $\mathcal{D}^{\text{OOD}}_{\text{val}}$, and the remaining portion was used as a test set $\mathcal{D}^{\text{OOD}}_{\text{tst}}$. The trained model was evaluated on $\mathcal{D}^{\text{ID}}_{\text{tst}}$ for ID classification accuracy and on both $\mathcal{D}^{\text{ID}}_{\text{tst}}$ and $\mathcal{D}^{\text{OOD}}_{\text{tst}}$ for OOD detection performance.

We investigated five NLL methods introduced in Section 3: SOP and ELR from the noise-robust regularization approach, and ProMix, PGDF, and TCL from the SSL-based approach. Additionally, a baseline method that trains a model using the standard cross-entropy loss without NLL was included for comparison. For OOD detection, we employed the 19 post-hoc methods listed in Table 1, as implemented in the OpenOOD benchmark.

Performance was evaluated using two perspectives. For ID classification, we used the classification accuracy on ID data. For OOD detection, we used two metrics: (1) the area under the receiver operating characteristic curve (AUROC) and (2) the area under the precision-recall curve (AUPR) with ID instances treated as positive. We also report the number of OOD detection methods for which each NLL configuration with our proposed propositions outperformed the baseline (No. Wins). These metrics were computed for each ID-OOD dataset pair and then averaged within the Near-OOD and Far-OOD groups.

The source code used in this study is publicly available at https://github.com/yujinlee815/robust-ood-noisy-labels. All experiments were

Table 1
Overview of the post-hoc OOD detection methods evaluated in the experiments.

Category	OOD Detection Methods
Softmax-based	MSP (Hendrycks & Gimpel, 2017), TempScale (Guo et al., 2017), ODIN (Liang et al., 2018)
Logit-based	MLS (Hendrycks et al., 2022), EBO (Liu et al., 2020b), ReAct (Sun et al., 2021), RankFeat (Song et al., 2022b), DICE (Sun & Li, 2022), ASH (Djurisic et al., 2023)
Distance-based	MDS (Lee et al., 2018), MDSEns (Lee et al., 2018), RMDS (Ren et al., 2021), KLM (Hendrycks et al., 2022), OpenMax (Bendale & Boult, 2016), SHE (Zhang et al., 2023a), Gram (Sastry & Oore, 2020), KNN (Sun et al., 2022), VIM (Wang et al., 2022)
Gradient-based	GradNorm (Huang et al., 2021)

repeated three times independently with different random seeds. The results reported are the averages and standard deviations over these three runs.

5.3. Implementation details for NLL methods

The baseline (w/o NLL) followed the configuration used in the OpenOOD benchmark with minor adjustments. For the five NLL methods (SOP, ELR, ProMix, PGDF, and TCL), we primarily followed the configurations of their respective official implementations described in Section 3, with minor modifications to improve OOD detection performance.

Below, we detail the specific modifications made to each method relative to its original setup. For all methods, a model with the ResNet-18 architecture was trained using the SGD optimizer with a momentum of 0.9 to ensure a fair and consistent comparison. The configurations used for Tiny-ImageNet were identical to those used for CIFAR-100 unless otherwise specified. For the baseline, the number of training epochs was increased to 300 for all datasets to ensure a fair comparison with the NLL methods. For SOP, we followed the hyperparameters of the symmetric noise setting except for the learning rate ratios. The learning rate ratios α_u and α_v were adjusted to 1 and 5 for CIFAR-100 and Tiny-ImageNet. For ELR, we also adopted the hyperparameters from the symmetric noise setting with a cosine annealing learning rate scheduler. For ProMix and PGDF, the number of training epochs was reduced by half, with the learning rate switch epochs additionally halved for PGDF. For TCL, the number of training epochs was set to 300 for Tiny-ImageNet.

Following Proposition 2, we applied energy-based sharpening with the hyperparameter $\gamma = 0.001$ for all SSL-based NLL methods (ProMix, PGDF, and TCL). Following Proposition 3, we modified NLL methods that originally used Mixup (ELR, ProMix, PGDF, and TCL) to exclude Mixup-related components from their training, as detailed in Section 4.2.

6. Results and discussion

6.1. Proposition 1. NLL enhances OOD detection under noisy labels

Tables 2 and 3 present the OOD detection performance of the proposed framework across various noisy scenarios in terms of AUROC and AUPR, respectively. Each NLL method (SOP, ELR, ProMix, PGDF, and TCL) incorporates Propositions 2 and 3.

Compared against the baseline (w/o NLL), the use of NLL consistently improved both OOD detection performance and ID classification accuracy. As the label noise ratio increased, the performance with NLL methods remained stable without significant degradation, while the baseline performance declined substantially. These results demonstrate the validity of Proposition 1 for robust OOD detection in severe noisy label scenarios. The performance gain was more pronounced in the Near-OOD setting, where the semantic proximity to ID instances makes detection more sensitive.

Fig. 5 presents the qualitative results of ID–OOD separability in terms of confidence score distributions and feature representations. Training without NLL methods blurred the distinction between ID and OOD in the presence of noisy labels, as shown in (b) and (f). By contrast, applying NLL preserved a clearer separation, with both ProMix and SOP incorporating Propositions 2 and 3 demonstrating patterns similar to those of the clean-label baseline in (a) and (e).

Further analysis reveals that among the NLL methods, SOP and ELR, which belong to the noise-robust regularization approach, exhibited the highest performance gains for distance-based OOD detection methods. ProMix, PGDF, and TCL, which belong to the SSL-based approach, tended to yield the highest gains for softmax-based, logit-based, and gradient-based OOD detection methods. Beyond noisy-label scenarios, NLL also improved OOD detection over the baseline even under the clean-label scenarios on CIFAR-10N and Tiny-ImageNet. Since these datasets are known to contain mislabeled or ambiguous instances (Northcutt et al., 2021), NLL helped mitigate their adverse effects.

It should be noted that naively applying NLL methods without Propositions 2 and 3 improves ID classification accuracy but does not always lead to meaningful improvements in OOD detection performance, as evident from Tables 4 and 5. This suggests that improvements in ID classification accuracy do not necessarily translate into better OOD detection performance, and that applying Propositions 2 and 3 is important when adopting NLL methods to improve OOD detection. Fig. 6–9 visualize their effects for ELR, ProMix, PGDF, and TCL, respectively.

6.2. Proposition 2. Sharpening model outputs in SSL-Based NLL benefits OOD detection

We evaluated the effect of applying model output sharpening to SSL-based NLL methods (ProMix, PGDF, and TCL) to demonstrate the validity of Proposition 2. Tables 4 and 5 show the effects of Propositions 2 and 3 on OOD detection performance in terms of AUROC and AUPR, respectively.

For ProMix, introducing Proposition 2 consistently yielded higher AUROC and AUPR than its original implementation across noisy label scenarios. The qualitative results in Fig. 7 further confirm this effectiveness. Model output sharpening produces a sharper confidence score distribution for ID instances with clearer ID-OOD separability in both the softmax output space and feature space.For TCL, Proposition 2 likewise improved performance over its original implementation across noisy label scenarios, as further corroborated by the qualitative results in Fig. 9.For PGDF, however, Proposition 2 alone did not improve OOD detection over its original implementation. Nevertheless, combining it with Proposition 3 resulted in a synergistic effect, achieving better OOD detection performance than applying Proposition 3 alone. These observations are also reflected in the qualitative results presented in Fig. 8.

Overall, Proposition 2 is more effective when combined with Proposition 3. We conjecture that applying model output sharpening to Mixup-augmented instances encourages the model to make overconfident predictions for uncertain instances, thereby hindering OOD detection performance.

6.3. Proposition 3. Mixup in NLL can degrade OOD detection despite ID accuracy gains

We evaluated the effect of removing Mixup-related components from NLL methods (ELR, ProMix, PGDF, and TCL) to demonstrate the validity of Proposition 3. The results are shown in Tables 4 and 5.

Introducing Proposition 3 improved OOD detection performance, albeit at the cost of reduced ID classification accuracy compared to the original implementations. This indicates that Mixup-related components, while beneficial for ID classification, may not always be

Table 2
Comparison of OOD detection performance in terms of AUROC across NLL methods (SOP, ELR, ProMix, PGDF, and TCL) under various noisy label scenarios. The best results are highlighted in **bold**, and the second-best results are underlined.

	Type	Metric		Baseline (w/o NLL)		SOP		ELR		ProMix		PGDF		TCL	
				Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD
CIFAR-10N	Clean (0%)	AUROC	Softmax-based	87.68±0.50	90.36±0.50	89.56±0.39	91.35±0.56	86.83±0.38	89.67±0.30	91.40±0.38	93.57±0.56	91.80±0.31	93.74±0.09	89.45±0.24	89.19±0.51
			Logit-based	85.67±0.77	87.74±1.97	86.85±1.66	88.89±1.14	86.13±0.19	88.55±0.34	91.11±0.39	95.21±0.42	89.99±0.48	94.14±0.29	83.55±1.30	88.42±1.89
			Distance-based	82.43±0.57	86.93±0.81	83.79±0.47	88.45±0.30	80.73±0.16	85.81±0.31	83.91±0.27	89.64±0.30	84.35±0.18	88.65±0.45	77.77±0.10	84.90±0.12
			Gradient-based	63.15±0.84	64.84±1.56	66.67±2.11	69.14±0.04	83.35±1.12	85.92±1.01	83.95±1.23	90.15±1.16	82.19±0.45	88.11±0.45	84.22±1.35	86.77±2.25
			Mean	83.26±0.58	86.56±1.15	84.77±0.89	88.03±0.52	83.54±0.23	87.29±0.18	**87.37±0.23**	**92.05±0.31**	87.20±0.16	91.16±0.28	81.78±0.47	86.78±0.60
		No. Wins		-	-	16	16	5	7	**17**	**18**	**17**	**18**	12	11
		ID Accuracy		95.62±0.01		**96.16±0.12**		93.68±0.22		94.24±0.07		95.54±0.15		91.72±0.82	
	Aggregate (9.03%)	AUROC	Softmax-based	83.24±0.33	87.10±0.41	89.04±0.39	91.86±0.09	86.30±0.16	89.16±0.41	91.15±0.10	93.52±0.09	85.78±0.93	86.99±1.45	89.97±0.35	90.36±0.40
			Logit-based	79.85±0.12	80.99±1.12	85.41±0.48	87.64±1.35	82.32±0.42	83.81±0.37	90.50±0.10	94.85±0.19	82.22±0.98	85.50±2.17	85.93±0.98	90.78±0.21
			Distance-based	76.29±0.37	81.52±0.51	82.41±0.66	87.33±0.83	79.52±0.22	84.81±0.55	83.53±0.22	89.22±0.22	78.24±0.66	82.68±0.67	78.65±0.51	85.43±0.39
			Gradient-based	65.87±0.52	65.05±1.58	65.77±5.00	71.29±5.27	78.53±0.43	80.91±0.46	81.62±2.47	88.53±1.73	76.79±1.60	82.45±4.15	85.18±1.53	86.89±1.47
			Mean	77.97±0.18	81.37±0.56	83.53±0.23	87.30±0.60	81.42±0.12	84.98±0.07	**86.83±0.27**	**91.64±0.10**	80.61±0.83	84.24±1.45	83.08±0.51	87.98±0.19
		No. Wins		–	–	12	13	16	16	**18**	**18**	15	13	16	**18**
		ID Accuracy		90.48±0.06		**95.70±0.31**		92.41±0.11		94.24±0.03		91.51±0.24		92.61±0.66	
	Random1 (17.23%)	AUROC	Softmax-based	76.52±3.84	78.23±5.12	88.42±0.37	91.08±0.20	87.64±0.14	90.36±0.47	91.32±0.22	93.46±0.44	85.01±0.37	86.50±1.10	89.82±0.19	90.06±0.19
			Logit-based	75.72±1.17	77.30±2.38	86.03±0.51	89.00±0.35	84.95±0.48	87.33±0.27	90.98±0.18	95.01±0.36	81.35±0.81	86.06±1.51	82.66±0.77	86.21±1.02
			Distance-based	70.88±0.84	75.46±1.11	83.11±0.44	88.14±0.30	81.07±0.30	85.38±0.37	83.64±0.11	89.30±0.12	76.53±0.39	81.13±1.08	78.51±0.13	85.46±0.19
			Gradient-based	73.43±2.02	74.46±4.65	64.74±3.00	69.81±2.73	84.40±0.31	87.44±0.62	83.23±1.10	89.14±1.74	74.94±1.56	79.81±1.30	82.97±1.12	84.08±1.69
			Mean	73.44±0.17	76.42±0.99	83.90±0.53	87.91±0.39	83.51±0.31	86.89±0.36	**87.15±0.10**	**91.75±0.23**	79.31±0.46	83.46±1.21	81.84±0.29	86.35±0.35
		No. Wins		–	–	17	18	18	**19**	**19**	**19**	18	**19**	17	18
		ID Accuracy		84.86±0.29		**95.33±0.10**		92.23±0.17		94.26±0.09		89.65±0.26		93.31±0.19	
	Worst (40.21%)	AUROC	Softmax-based	77.80±0.76	82.15±2.72	86.82±0.35	89.28±0.87	85.68±0.09	88.52±0.72	90.55±0.40	93.03±0.18	80.17±1.09	81.54±1.20	83.67±0.37	83.45±0.64
			Logit-based	75.09±1.07	80.92±1.81	84.54±0.91	87.60±1.36	82.88±0.37	84.34±0.84	89.60±0.76	94.33±0.32	77.54±0.65	81.72±0.63	77.29±1.45	81.92±0.82
			Distance-based	69.35±0.31	74.92±0.13	81.37±0.28	86.57±0.78	78.89±0.19	83.67±0.47	83.09±0.21	88.77±0.20	70.55±1.27	76.52±1.06	74.08±0.10	81.58±0.28
			Gradient-based	73.61±1.62	80.76±3.61	69.25±1.75	74.78±2.38	82.34±0.31	83.62±1.13	78.81±1.23	87.08±0.65	73.60±3.89	76.86±5.13	79.70±0.64	79.11±2.52
			Mean	72.72±0.68	78.26±1.13	82.59±0.35	86.70±1.03	81.40±0.20	84.65±0.65	**86.10±0.29**	**91.11±0.08**	74.44±0.46	78.97±0.20	76.90±0.38	81.85±0.14
		No. Wins		–	–	18	18	16	17	**19**	**19**	16	12	15	14
		ID Accuracy		78.31±0.94		92.21±0.62		87.52±0.13		**93.63±0.09**		79.87±0.78		85.17±0.75	
CIFAR-100N	Clean (0%)	AUROC	Softmax-based	81.45±0.14	80.09±0.78	81.14±0.13	75.35±1.27	78.31±0.16	78.50±0.44	79.92±0.41	71.83±0.83	80.92±0.21	77.31±1.22	77.28±0.20	69.78±1.43
			Logit-based	77.99±0.09	76.22±0.89	76.03±0.34	69.76±1.50	76.56±0.10	80.81±0.62	77.06±0.83	74.94±1.13	68.57±0.52	67.15±0.32	61.23±2.49	62.71±2.14
			Distance-based	70.41±0.09	76.58±0.43	69.88±0.06	75.86±0.73	68.87±0.12	76.10±0.84	69.42±0.12	74.27±0.13	69.49±0.01	78.24±0.86	65.20±0.44	72.23±0.64
			Gradient-based	78.10±0.21	75.93±1.53	76.52±0.46	62.16±1.80	66.67±0.98	70.46±2.91	66.43±0.41	58.84±2.47	65.44±0.43	56.05±1.76	72.62±0.57	74.24±0.76
			Mean	**74.95±0.05**	76.99±0.58	73.95±0.14	73.13±0.99	72.67±0.13	**77.67±0.22**	73.33±0.31	73.28±0.67	70.79±0.21	73.43±0.57	66.24±1.04	68.94±0.39
		No. Wins		–	–	**5**	4	4	**9**	4	4	4	6	2	2
		ID Accuracy		78.85±0.17		**79.00±0.08**		75.56±0.11		74.10±0.17		77.67±0.22		72.21±0.29	

Table 2
Continued.

	Noisy (40.20%)	AUROC	Softmax-based	71.49±0.36	73.84±0.22	77.00±0.07	73.68±0.74	74.69±0.26	76.17±1.00	77.68±0.12	71.79±1.41	71.42±1.23	66.82±1.36	72.73±0.11	67.09±0.99
			Logit-based	68.75±0.28	72.55±0.63	73.55±0.21	68.69±0.76	69.61±0.60	73.62±0.37	75.40±0.17	75.61±1.27	67.90±0.77	66.60±1.39	62.55±2.57	64.23±0.65
			Distance-based	61.50±0.20	70.23±0.60	66.84±0.03	74.69±0.55	63.43±0.32	70.25±0.38	66.93±0.15	73.47±0.75	60.38±0.45	70.65±0.78	62.05±0.20	69.71±0.48
			Gradient-based	68.40±0.72	70.05±0.25	73.05±0.33	62.88±1.44	43.70±0.75	40.99±1.60	65.03±0.82	63.37±1.27	71.10±0.33	66.32±2.74	69.56±0.35	71.15±4.57
			Mean	65.73±0.12	71.52±0.50	70.89±0.07	72.02±0.42	66.12±0.41	70.71±0.36	**71.20±0.13**	**73.35±0.96**	65.06±0.58	68.54±1.13	64.29±0.91	67.64±0.13
		No. Wins		–	–	**17**	9	13	12	**17**	**13**	8	4	11	7
		ID Accuracy		53.91±0.46		66.39±0.24		62.50±0.15		**66.91±0.17**		60.25±0.31		62.99±0.38	
Tiny-ImageNet	Clean (0%)	AUROC	Softmax-based	69.82±0.27	70.63±0.54	71.81±0.44	70.46±0.39	75.41±0.13	77.03±0.56	70.87±0.43	69.64±0.49	72.38±0.24	73.86±0.66	68.77±0.43	67.79±0.63
			Logit-based	66.74±0.13	67.76±0.20	68.85±1.60	69.09±3.76	68.08±0.67	66.40±1.54	68.15±1.44	68.20±1.47	70.07±1.89	72.92±4.03	62.21±0.64	63.15±1.10
			Distance-based	67.08±0.12	74.27±0.51	73.61±1.67	78.24±1.57	68.85±0.19	72.97±0.47	69.90±0.28	74.74±0.49	68.32±0.35	73.86±0.98	69.91±1.93	73.58±1.32
			Gradient-based	68.52±0.52	71.46±0.25	72.67±3.56	71.50±4.62	75.32±0.18	86.17±0.53	65.51±1.42	68.53±0.57	71.41±1.05	77.80±0.38	76.34±0.28	73.40±1.17
			Mean	67.48±0.08	71.49±0.34	**71.77±1.14**	**73.77±2.10**	69.98±0.19	72.23±0.76	69.27±0.52	71.54±0.63	69.68±0.68	**73.77±1.81**	67.64±0.91	69.36±0.81
		No. Wins		–	–	15	13	**16**	**14**	**16**	5	15	**14**	11	6
		ID Accuracy		62.45±0.28		**67.70±0.35**		67.54±0.29		65.16±0.04		67.13±0.40		57.26±0.47	
	Sym0.2 (20%)	AUROC	Softmax-based	63.32±1.26	61.97±1.74	70.78±0.67	69.01±0.59	73.08±0.19	72.36±0.32	70.34±0.27	70.06±0.66	67.44±0.96	69.32±0.35	65.33±0.73	64.65±0.48
			Logit-based	62.49±1.95	65.61±2.03	64.55±1.75	61.41±2.20	64.72±0.05	62.00±0.30	66.61±0.48	65.92±1.32	64.84±0.88	68.18±0.49	67.33±2.31	69.46±1.56
			Distance-based	64.43±0.30	70.94±0.87	72.30±1.69	77.53±1.84	67.46±0.22	70.90±0.58	69.51±0.23	74.68±0.53	65.53±1.65	71.16±1.64	67.55±0.84	72.97±0.74
			Gradient-based	65.69±2.11	69.23±0.45	71.58±2.04	69.15±2.52	69.81±0.43	72.44±1.11	65.34±0.94	69.10±1.18	68.25±1.18	75.56±2.08	74.80±2.11	73.18±1.93
			Mean	63.71±0.86	67.75±0.82	**69.58±0.25**	70.65±0.47	67.61±0.10	68.40±0.30	68.50±0.19	**70.89±0.47**	65.76±1.05	70.16±0.66	67.51±1.20	70.56±0.82
		No. Wins		–	–	17	15	15	13	**18**	**17**	15	15	16	14
		ID Accuracy		48.82±0.44		65.37±0.19		63.93±0.13		**65.51±0.17**		60.86±1.01		61.41±0.02	
	Sym0.5 (50%)	AUROC	Softmax-based	59.84±1.07	58.51±0.71	69.40±0.48	67.89±0.79	70.73±1.20	68.33±1.69	67.13±0.52	66.71±0.38	64.51±1.04	64.21±1.37	61.09±0.58	59.55±0.53
			Logit-based	57.58±0.56	60.44±0.95	62.36±1.83	61.33±1.23	63.49±2.08	60.39±2.61	64.23±0.55	62.59±1.12	69.89±2.26	73.59±1.64	66.93±2.21	69.48±1.28
			Distance-based	60.17±0.51	66.52±0.82	68.43±0.56	72.03±1.19	65.02±0.12	69.18±0.41	68.14±0.24	73.60±0.49	63.09±2.22	69.22±2.61	66.27±0.69	71.84±0.69
			Gradient-based	61.43±1.69	64.91±1.64	71.56±2.15	69.91±0.78	70.28±1.55	68.38±2.92	64.86±0.90	68.56±2.38	70.21±3.82	72.47±4.14	75.18±5.72	71.53±6.07
			Mean	59.37±0.65	63.25±0.31	**66.83±0.62**	67.89±1.04	65.71±0.86	66.23±1.19	66.58±0.16	68.77±0.48	65.84±0.55	**69.98±0.51**	66.13±0.72	69.14±0.54
		No. Wins		–	–	15	15	16	15	**18**	**17**	16	16	15	16
		ID Accuracy		38.98±0.32		59.75±0.12		55.64±0.56		**60.66±0.23**		51.52±1.97		54.86±0.39	
	Asym0.4 (40%)	AUROC	Softmax-based	68.90±0.94	68.46±0.66	70.53±0.93	68.80±0.45	75.19±0.08	76.51±0.62	70.08±0.46	68.36±0.62	69.05±0.21	70.23±0.67	64.15±0.64	62.13±1.08
			Logit-based	67.02±0.69	68.21±0.90	69.33±1.02	70.44±0.63	67.61±1.22	65.77±1.93	67.16±1.05	66.26±1.19	67.86±1.53	73.02±1.15	71.20±2.40	72.33±2.21
			Distance-based	66.76±0.27	73.80±0.27	71.80±0.80	76.91±1.02	68.37±0.25	72.66±0.17	68.99±0.29	73.58±0.38	69.87±0.24	73.83±0.50	67.40±1.46	72.07±1.29
			Gradient-based	68.69±1.31	71.97±0.16	71.48±1.61	70.97±0.95	75.39±0.69	85.03±0.48	66.54±1.90	69.67±1.56	65.79±1.08	73.76±0.28	80.96±0.96	79.24±0.99
			Mean	67.28±0.37	71.10±0.44	**70.80±0.23**	**73.27±0.60**	69.58±0.44	71.75±0.69	68.45±0.39	70.24±0.31	68.89±0.68	73.00±0.50	68.80±1.37	70.96±1.19
		No. Wins		–	–	15	12	**16**	13	14	9	12	**15**	9	8
		ID Accuracy		57.44±0.43		62.99±0.39		**66.27±0.33**		63.62±0.13		64.34±0.45		57.26±0.47	

Table 3
Comparison of OOD detection performance in terms of AUPR across NLL methods (SOP, ELR, ProMix, PGDF, and TCL) under various noisy label scenarios. The best results are highlighted in **bold**, and the second-best results are underlined.

Dataset	Type	Metric		Baseline (w/o NLL)		SOP		ELR		ProMix		PGDF		TCL	
				Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD
CIFAR-10N	Clean (0%)	AUPR	Softmax-based	85.55±0.71	73.49±2.06	87.47±0.55	73.97±0.95	86.94±0.76	74.38±2.37	92.85±0.40	89.42±1.02	92.79±0.34	87.65±0.21	91.42±0.27	79.20±1.88
			Logit-based	83.86±0.73	71.60±3.21	84.23±1.94	70.28±1.86	86.41±0.62	73.72±1.51	92.07±0.44	90.93±0.85	90.51±0.51	87.51±0.66	86.26±1.11	80.63±2.64
			Distance-based	81.50±0.49	74.25±1.15	82.86±0.53	76.44±0.20	80.12±0.35	70.91±1.06	84.15±0.27	81.21±0.64	84.51±0.32	78.85±0.92	78.62±0.09	73.72±0.14
			Gradient-based	59.66±0.44	37.61±1.23	58.63±1.63	38.29±1.87	82.90±2.36	71.75±4.22	82.17±2.00	80.83±2.69	77.94±0.79	73.12±0.39	87.00±1.08	79.29±3.08
			Mean	81.73±0.46	71.36±1.86	82.74±1.03	72.10±0.60	83.33±0.59	72.39±1.53	**87.92±0.35**	**85.55±0.74**	87.37±0.17	82.67±0.45	83.49±0.37	77.06±0.69
		No. Wins		–	–	15	12	5	8	**17**	**18**	**17**	**18**	15	16
		ID Accuracy		95.62±0.01		**96.16±0.12**		93.68±0.22		94.24±0.07		95.54±0.15		91.72±0.82	
	Aggregate (9.03%)	AUPR	Softmax-based	82.00±0.53	71.56±2.22	87.32±1.04	76.25±0.49	85.90±0.22	73.31±1.65	92.56±0.15	89.32±0.16	87.27±0.88	73.63±3.51	91.82±0.43	81.73±0.64
			Logit-based	79.24±0.10	63.71±1.59	84.01±1.16	72.25±0.65	81.65±0.31	67.60±0.23	91.32±0.20	90.42±0.45	83.96±0.91	71.83±4.81	88.13±0.96	83.78±0.94
			Distance-based	76.28±0.44	67.18±0.50	81.67±0.67	74.92±1.98	78.30±0.48	70.10±1.78	83.75±0.34	80.87±0.24	78.55±0.72	67.68±1.86	79.43±0.50	74.77±0.39
			Gradient-based	62.38±0.45	38.05±2.63	61.03±6.52	44.21±7.19	74.27±0.68	59.27±0.91	79.31±3.31	79.19±3.26	74.35±2.28	62.97±9.27	87.94±1.17	80.14±1.72
			Mean	77.39±0.21	65.24±1.14	82.21±0.55	72.67±0.68	80.35±0.21	69.25±0.69	**87.30±0.39**	**85.13±0.25**	81.41±0.83	69.68±3.33	84.58±0.54	79.00±0.37
		No. Wins		–	–	12	13	13	13	**18**	**19**	15	14	17	18
		ID Accuracy		90.48±0.06		**95.70±0.31**		92.41±0.11		94.24±0.03		91.51±0.24		92.61±0.66	
	Random1 (17.23%)	AUPR	Softmax-based	77.67±3.81	62.02±5.59	86.01±0.69	74.14±0.79	87.73±0.17	75.78±1.43	92.82±0.25	89.30±0.78	86.66±0.38	72.75±3.05	91.56±0.26	81.28±0.87
			Logit-based	76.59±1.60	61.14±3.77	83.18±0.64	70.75±1.22	85.34±0.26	74.42±0.87	91.97±0.24	90.75±0.70	82.63±0.94	71.42±3.23	85.50±0.68	76.96±1.43
			Distance-based	71.74±0.91	60.68±2.55	82.20±0.49	76.20±0.45	80.64±0.54	71.01±0.06	83.97±0.04	80.74±0.21	76.79±0.25	65.17±1.85	79.25±0.21	74.66±0.32
			Gradient-based	72.18±3.19	54.93±4.74	57.38±2.19	39.77±3.22	82.38±0.22	72.95±0.70	81.68±1.94	79.63±3.38	71.45±2.24	56.65±2.74	86.22±0.83	75.43±2.39
			Mean	74.23±0.44	60.73±2.68	81.80±0.63	72.24±0.83	83.33±0.34	72.94±0.49	**87.77±0.20**	**85.20±0.58**	79.91±0.46	67.89±2.48	83.53±0.31	76.47±0.64
		No. Wins		–	–	17	16	17	**19**	**19**	**19**	17	18	18	18
		ID Accuracy		84.86±0.29		**95.33±0.10**		92.23±0.17		94.26±0.09		89.65±0.26		93.31±0.19	
	Worst (40.21%)	AUPR	Softmax-based	79.37±0.70	66.94±3.09	85.87±0.58	74.90±0.76	87.43±0.17	78.05±0.87	92.13±0.57	88.28±0.47	82.12±0.90	65.55±1.23	86.30±0.54	69.70±1.16
			Logit-based	76.58±1.05	66.28±1.66	82.68±1.18	70.67±2.01	84.62±0.37	71.63±1.04	90.50±0.92	89.28±0.45	78.90±0.36	65.10±1.69	80.37±1.33	69.69±0.16
			Distance-based	70.66±0.35	60.18±0.40	80.88±0.21	73.71±1.29	79.14±0.15	70.43±0.58	83.36±0.28	79.29±0.14	71.43±1.09	59.93±1.28	75.07±0.13	68.61±0.72
			Gradient-based	74.27±0.96	63.28±3.66	62.59±1.94	47.62±3.36	81.42±0.43	66.16±0.44	76.79±1.67	75.35±0.99	71.41±4.23	53.23±7.74	82.93±0.74	67.43±3.01
			Mean	74.09±0.63	63.34±1.19	81.27±0.52	71.57±1.50	82.30±0.18	71.78±0.71	**86.65±0.53**	**83.66±0.21**	75.48±0.25	62.10±1.00	78.93±0.31	69.06±0.64
		No. Wins		–	–	18	16	17	16	**19**	**19**	15	9	15	14
		ID Accuracy		78.31±0.94		92.21±0.62		87.52±0.13		**93.63±0.09**		79.87±0.78		85.17±0.75	
CIFAR-100N	Clean (0%)	AUPR	Softmax-based	84.30±0.09	68.49±1.44	83.90±0.10	62.51±2.25	81.48±0.21	67.85±1.32	83.83±0.40	60.27±1.65	83.84±0.44	65.22±2.37	80.85±0.27	55.17±1.66
			Logit-based	80.90±0.07	63.84±1.55	79.02±0.23	55.78±1.66	79.39±0.15	70.37±1.31	81.10±0.82	63.03±1.54	72.13±0.71	54.29±0.81	65.48±1.75	48.79±2.78
			Distance-based	73.86±0.07	64.03±0.48	72.62±0.04	61.73±1.28	72.09±0.23	63.14±0.87	72.86±0.24	62.26±0.58	71.69±0.48	64.69±1.58	68.72±0.41	59.22±0.57
			Gradient-based	77.69±0.30	54.01±1.62	76.55±0.36	37.80±0.71	70.75±1.07	56.22±4.21	67.69±0.16	40.65±1.68	65.16±0.34	35.19±2.21	78.13±0.69	65.02±2.18
			Mean	**77.93±0.04**	64.15±0.85	76.63±0.05	58.72±1.37	75.81±0.22	**65.80±0.73**	76.92±0.27	61.05±1.09	73.41±0.51	59.93±1.36	70.11±0.79	55.59±0.52
		No. Wins		–	–	4	2	**6**	**12**	**6**	6	3	6	3	5
		ID Accuracy		78.85±0.17		**79.00±0.08**		75.56±0.11		74.10±0.17		77.67±0.22		72.21±0.29	

Table 3
Continued.

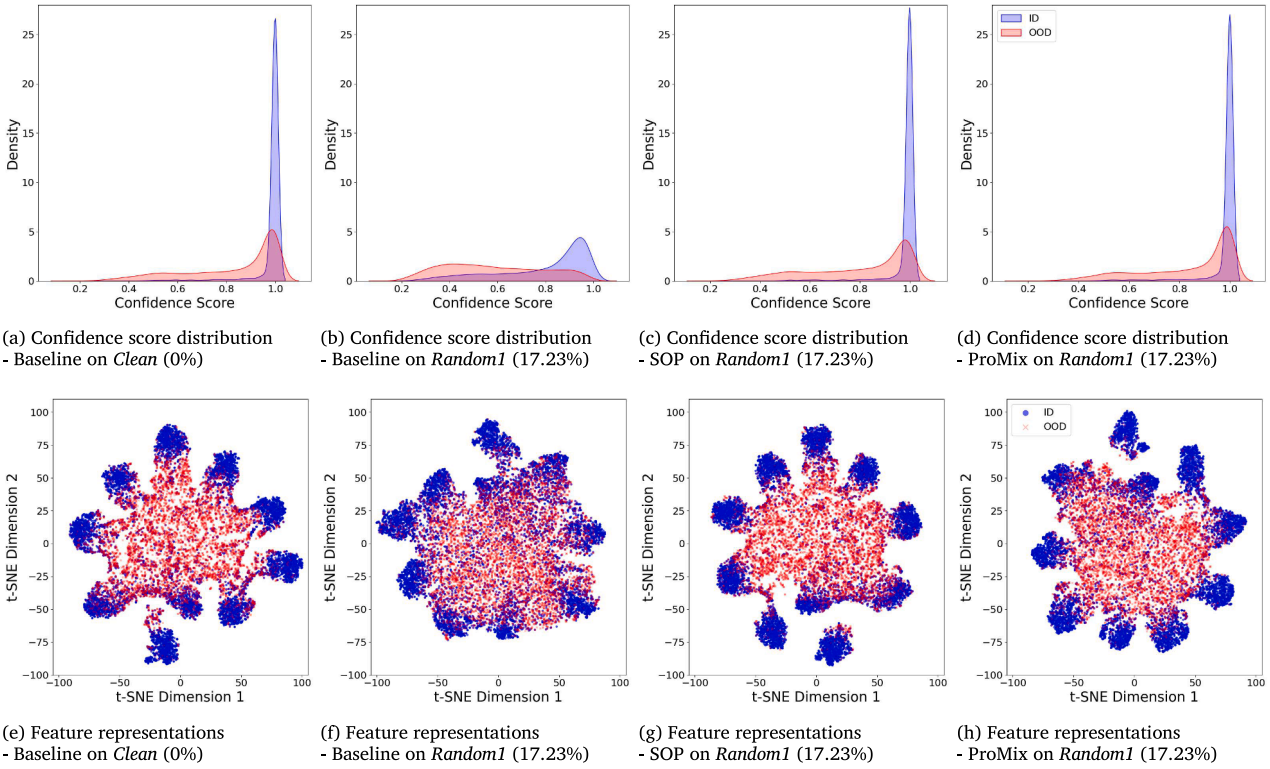
	Noisy (40.20%)	AUPR	Softmax-based	74.43±0.28	60.10±0.78	79.80±0.18	59.47±1.41	76.23±0.67	64.43±1.51	81.71±0.13	59.59±2.42	76.07±0.20	54.10±3.10	74.81±0.22	50.15±1.55
			Logit-based	71.72±0.27	58.02±1.33	76.46±0.32	53.59±0.91	71.45±0.83	60.62±0.77	79.53±0.11	61.82±1.86	71.83±0.58	52.26±2.34	66.20±2.71	48.93±0.31
			Distance-based	65.46±0.15	55.76±0.57	69.58±0.19	60.12±0.84	66.25±0.29	55.51±0.98	70.56±0.25	59.92±1.06	64.29±0.36	54.65±1.14	65.67±0.31	55.25±0.40
			Gradient-based	69.43±0.71	48.71±0.88	73.17±0.36	37.48±0.33	48.26±0.60	26.76±0.67	67.01±0.54	44.56±1.12	74.92±0.33	50.40±2.44	74.16±0.41	60.59±5.76
			Mean	69.06±0.12	56.79±0.82	73.55±0.19	56.76±0.78	68.52±0.52	57.02±0.91	**74.96±0.18**	**59.66±1.46**	69.09±0.33	53.58±1.84	67.73±1.04	52.73±0.11
		No. Wins		–	–	**17**	9	12	13	**17**	**16**	10	5	11	8
		ID Accuracy		53.91±0.46		66.39±0.24		62.50±0.15		**66.91±0.17**		60.25±0.31		62.99±0.38	
	Clean (0%)	AUPR	Softmax-based	58.92±0.21	73.38±0.45	64.00±0.21	74.84±0.40	65.78±0.13	79.43±0.55	61.50±0.53	73.17±0.43	61.53±0.72	76.12±0.55	59.97±0.15	72.18±0.58
			Logit-based	53.78±0.36	68.77±0.19	59.86±1.29	72.96±3.34	59.55±0.50	72.14±1.06	57.06±1.43	70.97±1.26	57.45±0.97	73.48±3.08	50.75±0.54	64.13±0.96
			Distance-based	54.13±0.10	73.27±0.48	62.13±1.61	78.38±1.45	57.74±0.20	73.62±0.51	57.79±0.36	74.70±0.49	55.87±0.07	74.02±0.98	58.16±1.92	73.69±1.59
			Gradient-based	52.53±0.84	68.76±0.69	62.01±5.88	73.66±6.64	59.54±0.05	83.68±0.55	49.60±1.45	66.31±0.37	57.88±1.33	77.26±0.65	64.55±0.06	73.37±1.15
			Mean	54.69±0.22	71.63±0.18	**61.70±1.39**	**75.86±2.09**	59.68±0.14	74.60±0.62	57.71±0.73	72.84±0.64	57.37±0.18	74.35±1.48	56.44±0.95	70.42±0.88
		No. Wins		–	–	**18**	**16**	16	15	**18**	12	16	**16**	13	8
		ID Accuracy		62.45±0.28		**67.70±0.35**		67.54±0.29		65.16±0.04		67.13±0.40		59.31±0.14	
	Sym0.2 (20%)	AUPR	Softmax-based	50.75±1.19	62.18±2.26	62.74±0.37	73.60±0.54	63.58±0.31	75.53±0.44	60.85±0.33	73.36±0.61	56.71±1.59	71.48±0.09	54.87±0.67	67.86±0.54
			Logit-based	48.47±1.40	63.26±1.80	57.07±1.10	66.88±1.71	56.60±0.29	68.14±0.43	55.29±0.63	68.96±1.21	52.93±1.38	69.01±0.36	53.54±1.95	68.53±1.42
			Distance-based	50.88±0.19	68.54±0.64	60.92±1.51	77.44±1.56	56.01±0.22	71.56±0.73	57.34±0.06	74.86±0.43	52.81±1.65	70.91±1.54	55.26±0.73	72.27±0.76
			Gradient-based	50.13±1.33	65.15±0.42	62.45±4.57	72.45±4.64	53.63±0.57	70.29±1.31	48.82±0.90	65.53±1.36	53.91±1.16	73.73±1.10	61.48±1.97	72.43±1.86
Tiny-ImageNet			Mean	50.06±0.78	65.69±0.95	**60.07±0.54**	**73.24±0.48**	57.27±0.24	71.04±0.53	56.80±0.26	72.27±0.50	53.52±1.43	70.55±0.71	54.98±1.01	70.40±0.81
		No. Wins		–	–	**18**	**18**	16	15	17	**18**	16	15	16	15
		ID Accuracy		48.82±0.44		65.37±0.19		63.93±0.13		**65.51±0.17**		60.86±1.01		61.41±0.02	
	Sym0.5 (50%)	AUPR	Softmax-based	47.58±0.92	59.12±0.91	60.50±0.72	71.95±0.69	61.12±1.06	71.72±1.35	58.23±0.67	70.31±0.52	53.29±1.43	66.02±1.37	51.01±0.63	63.13±0.31
			Logit-based	44.44±0.61	57.73±0.86	53.41±1.96	65.04±1.94	53.43±1.01	64.59±1.39	53.74±0.31	65.95±0.60	55.70±2.15	71.59±1.45	53.32±2.27	68.75±1.41
			Distance-based	47.29±0.78	64.11±1.09	57.70±0.11	72.89±0.79	53.10±0.03	69.45±0.27	55.87±0.28	73.12±0.56	51.31±1.26	68.49±2.03	54.67±0.53	71.08±0.70
			Gradient-based	47.20±1.70	60.43±1.74	62.98±4.18	74.22±2.50	55.77±1.70	68.69±2.79	49.26±0.62	64.87±2.28	56.42±2.91	70.69±3.16	62.55±6.32	69.57±6.10
			Mean	46.43±0.78	61.11±0.70	**57.07±0.45**	**70.33±0.98**	54.61±0.57	68.24±0.78	55.22±0.19	69.98±0.31	53.28±0.86	69.20±0.64	54.08±0.94	69.01±0.55
		No. Wins		–	–	17	17	16	17	**18**	**18**	**18**	16	16	16
		ID Accuracy		38.98±0.32		59.75±0.12		55.64±0.56		**60.66±0.23**		51.52±1.97		54.86±0.39	
	Asym0.4 (40%)	AUPR	Softmax-based	57.05±0.85	70.80±0.66	62.54±0.91	73.17±0.74	65.32±0.15	79.05±0.53	60.67±0.44	71.82±0.54	59.97±0.26	73.75±0.72	54.10±0.80	65.99±0.75
			Logit-based	53.70±0.70	68.91±0.75	60.07±0.79	73.93±0.69	58.98±0.54	71.39±1.00	56.23±0.76	69.47±0.95	57.65±1.07	74.71±1.23	58.52±2.38	72.93±2.59
			Distance-based	53.77±0.22	72.52±0.43	59.97±0.56	76.76±0.99	57.34±0.14	73.45±0.13	57.03±0.18	73.44±0.43	57.55±0.05	74.17±0.48	55.44±1.45	71.80±1.57
			Gradient-based	52.45±1.31	68.48±0.11	61.35±3.02	74.95±1.21	60.26±0.92	82.94±0.65	50.83±2.11	67.44±1.86	52.40±0.78	74.31±0.66	68.53±0.84	78.77±1.12
			Mean	54.19±0.46	70.90±0.51	**60.48±0.44**	**75.20±0.78**	59.27±0.19	74.18±0.40	57.03±0.46	71.61±0.26	57.69±0.37	74.28±0.57	56.89±1.32	71.61±1.47
		No. Wins		–	–	**18**	15	16	15	16	14	15	**16**	12	9
		ID Accuracy		57.44±0.43		62.99±0.39		**66.27±0.33**		63.62±0.13		64.34±0.45		57.26±0.47	

Table 4
Effects of Propositions 2 and 3 on OOD detection performance in terms of AUROC across NLL methods (ELR, ProMix, PGDF, and TCL). The best results are highlighted in **bold**, and the second-best results are underlined.

Dataset	Type	Metric	Baseline (w/o NLL)		ELR				ProMix								PGDF								TCL							
					ELR+		w/P3		Original		w/P2		w/P3		w/P2&P3		Original		w/P2		w/P3		w/P2&P3		Original		w/P2		w/P3		w/P2&P3	
			Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD
CIFAR-10N	Clean (0%)	AUROC Softmax-based	87.68±0.50	90.36±0.50	81.85±0.78	85.61±1.87	86.83±0.38	89.67±0.30	85.36±0.66	79.31±1.18	86.11±1.47	84.21±2.01	89.02±0.41	91.64±0.45	91.40±0.38	93.57±0.56	65.64±7.84	72.61±8.54	73.36±8.53	79.64±7.79	89.24±0.14	91.34±0.35	91.80±0.31	93.74±0.09	88.93±0.23	88.44±0.23	89.86±0.08	89.82±0.50	89.43±0.16	89.43±0.30	89.45±0.24	89.19±0.51
		AUROC Logit-based	85.67±0.77	87.74±1.97	82.84±0.68	86.04±1.40	86.13±0.19	88.55±0.34	79.63±0.37	73.43±0.84	80.31±2.73	79.04±2.09	86.74±0.20	91.49±0.46	91.11±0.39	95.21±0.42	68.90±6.79	76.68±6.55	53.14±1.74	67.28±1.67	83.91±1.61	87.98±1.26	89.99±0.48	94.14±0.29	78.20±0.84	81.48±1.61	81.98±0.71	85.12±0.62	80.31±1.34	86.61±1.91	83.55±1.30	88.42±1.89
		AUROC Distance-based	82.43±0.57	86.93±0.81	78.14±0.40	83.92±1.08	80.73±0.16	85.81±0.31	83.36±0.78	85.65±0.37	83.57±0.39	87.86±1.35	82.06±0.23	88.16±0.18	83.91±0.27	89.64±0.30	75.68±3.58	83.13±3.32	70.53±0.70	79.68±0.21	82.74±0.25	87.34±0.33	84.35±0.18	88.65±0.45	78.84±0.27	84.74±0.11	79.59±0.21	86.50±0.49	77.37±0.35	84.99±0.41	77.77±0.10	84.90±0.12
		AUROC Gradient-based	63.15±0.84	64.84±1.56	77.86±1.03	81.11±2.64	83.35±1.12	85.92±1.01	80.75±1.96	69.26±1.66	82.33±3.21	79.61±2.09	56.22±2.11	71.01±3.29	83.95±1.23	90.15±1.16	47.35±13.07	59.72±12.16	48.62±1.91	63.94±2.59	61.18±0.25	68.94±1.33	82.19±0.45	88.11±0.45	86.02±0.93	87.64±0.57	83.60±2.29	86.52±1.22	83.03±2.90	84.80±1.09	84.22±1.35	86.77±2.25
		Mean	83.26±0.58	86.56±1.15	80.20±0.53	84.71±1.16	**83.54±0.23**	**87.29±0.18**	82.36±0.58	79.93±0.19	82.88±1.12	84.06±1.03	83.27±0.15	88.86±0.38	**87.20±0.16**	**91.16±0.28**	70.46±4.56	78.20±4.58	64.33±0.60	74.93±0.57	83.00±0.63	87.21±0.59	**87.20±0.16**	**91.16±0.28**	80.61±0.31	84.45±0.62	**82.18±0.06**	86.59±0.52	80.50±0.44	86.20±0.73	81.78±0.47	**86.78±0.60**
		ID Accuracy	95.62±0.01		**94.15±0.08**		93.68±0.22		96.05±0.01		**96.99±0.08**		94.18±0.15		94.24±0.07		95.68±0.15		**95.97±0.10**		95.46±0.21		95.54±0.15		**94.39±0.16**		94.31±0.23		91.83±0.88		91.72±0.82	
	Aggregate (9.03%)	AUROC Softmax-based	83.24±0.33	87.10±0.41	72.22±2.29	73.49±3.60	86.30±0.16	89.16±0.41	85.51±0.08	80.87±3.03	86.85±0.65	83.04±2.90	88.85±0.19	90.83±0.24	91.15±0.10	93.52±0.09	75.07±9.61	81.31±7.00	80.38±1.48	85.50±0.76	84.47±0.73	85.11±0.81	85.78±0.93	86.99±1.45	88.35±0.70	87.70±2.46	88.39±0.28	87.72±1.49	89.97±0.19	89.91±0.32	89.97±0.35	90.36±0.40
		AUROC Logit-based	79.85±0.12	80.99±1.12	73.57±1.66	75.15±0.95	82.32±0.42	83.81±0.37	79.92±0.95	76.42±5.64	81.01±3.32	77.83±4.81	86.69±0.20	90.99±0.35	90.50±0.10	94.85±0.19	72.75±1.51	82.20±0.90	60.70±1.20	72.91±1.45	81.43±1.34	83.67±1.07	82.22±0.98	85.50±2.17	79.70±0.25	81.79±1.09	79.76±2.01	82.74±0.44	82.90±1.04	88.74±1.50	85.93±0.98	90.78±0.21
		AUROC Distance-based	76.29±0.37	81.52±0.51	75.41±0.58	80.60±0.45	79.52±0.22	84.81±0.55	84.02±0.19	88.09±1.90	81.98±1.41	85.93±1.94	82.62±0.18	88.00±0.42	83.53±0.22	89.22±0.22	76.62±0.68	83.11±0.36	72.47±0.93	79.84±0.31	78.45±0.39	82.81±0.53	78.24±0.66	82.68±0.67	79.05±0.34	85.40±0.81	78.97±0.19	85.27±0.07	77.86±0.32	85.13±0.10	78.65±0.51	85.43±0.39
		AUROC Gradient-based	65.87±0.52	65.05±1.58	69.29±3.02	71.47±3.21	78.53±0.43	80.91±0.46	79.45±2.06	74.63±7.45	82.51±2.38	76.07±6.63	57.77±0.79	71.43±1.18	81.62±2.47	88.53±1.73	56.67±3.31	70.90±3.24	58.24±1.48	71.47±1.78	65.05±2.22	71.07±1.33	76.79±1.60	82.45±4.15	85.58±1.91	86.23±3.57	85.19±1.85	86.04±1.96	84.63±2.18	86.76±1.12	85.18±1.53	86.89±1.47
		Mean	77.97±0.18	81.37±0.56	74.00±1.24	77.27±1.11	**81.42±0.12**	**84.98±0.07**	82.72±0.36	82.56±3.48	82.47±1.91	82.40±3.13	83.58±0.18	88.52±0.40	**86.83±0.27**	**91.64±0.10**	74.10±2.18	81.89±1.35	69.25±0.23	78.10±0.49	79.64±0.78	82.82±0.31	**80.61±0.83**	**84.24±1.45**	81.07±0.29	84.67±1.28	81.03±0.60	84.90±0.16	81.72±0.14	87.11±0.53	**83.08±0.51**	**87.98±0.19**
		ID Accuracy	90.48±0.06		**93.27±0.02**		92.41±0.11		96.46±0.48		**96.83±0.05**		94.06±0.16		94.24±0.03		94.06±0.05		**94.14±0.22**		91.71±0.23		91.51±0.24		**94.71±0.28**		94.68±0.02		92.70±0.59		92.61±0.66	
	Random1 (17.23%)	AUROC Softmax-based	76.52±3.84	78.23±5.12	72.95±1.57	75.75±1.65	87.64±0.14	90.36±0.47	85.57±1.48	80.63±0.83	86.47±0.36	81.21±2.78	89.04±0.20	91.62±0.09	91.32±0.22	93.46±0.44	64.47±1.12	70.95±2.69	82.99±0.65	86.94±0.91	84.34±0.68	86.11±0.79	85.01±0.37	86.50±1.10	86.60±0.97	84.86±1.62	86.74±0.39	84.00±2.48	89.63±0.18	89.14±0.32	89.82±0.19	90.06±0.19
		AUROC Logit-based	75.72±1.17	77.30±2.38	73.47±1.07	77.68±1.32	84.95±0.48	87.33±0.27	80.54±2.52	76.39±0.43	80.17±2.50	75.25±5.13	86.63±0.41	91.46±0.07	90.98±0.18	95.01±0.36	74.02±1.86	81.17±2.44	60.30±3.46	73.47±2.68	80.42±0.99	84.88±1.04	81.35±0.81	86.06±1.51	74.48±2.06	77.56±2.32	81.90±2.97	84.22±2.08	82.82±1.40	86.28±3.23	82.66±0.77	86.21±1.02
		AUROC Distance-based	70.88±0.84	75.46±1.11	76.04±0.51	80.70±0.53	81.07±0.30	85.38±0.37	83.07±0.15	86.37±1.32	82.97±1.09	86.65±2.03	82.19±0.44	88.22±0.30	83.64±0.11	89.30±0.12	76.17±0.37	82.98±1.18	72.45±0.12	79.66±0.98	76.60±0.44	82.10±0.49	76.53±0.39	81.13±1.08	77.58±0.47	84.56±0.62	78.42±0.31	84.61±0.18	78.27±0.35	84.89±0.59	78.51±0.13	85.46±0.19
		AUROC Gradient-based	73.43±2.02	74.46±4.65	62.08±1.78	65.39±2.94	84.40±0.31	87.44±0.62	77.14±1.04	67.92±5.22	83.55±3.43	76.20±7.67	57.72±1.15	72.17±0.58	83.23±1.10	89.14±1.74	64.25±2.44	71.92±3.08	63.19±1.68	75.64±3.23	66.44±1.05	73.89±2.14	74.94±1.56	79.81±1.30	82.52±1.91	83.48±1.72	83.54±1.14	85.81±2.75	84.93±0.83	86.51±0.85	82.97±1.12	84.08±1.69
		Mean	73.44±0.17	76.42±0.99	74.00±0.72	78.16±0.90	**83.51±0.31**	**86.89±0.36**	82.36±1.00	81.34±0.64	82.67±1.51	81.64±3.36	83.39±0.36	88.94±0.16	**87.15±0.10**	**91.75±0.23**	73.02±0.98	79.93±1.84	69.79±1.30	78.64±1.43	78.49±0.68	83.18±0.75	**79.31±0.46**	**83.46±1.21**	78.29±0.84	82.34±0.45	81.10±1.05	84.46±0.86	**81.85±0.66**	86.09±1.38	81.84±0.29	**86.35±0.35**
		ID Accuracy	84.86±0.29		**92.74±0.05**		92.23±0.17		96.37±0.22		**96.48±0.13**		94.00±0.05		94.26±0.09		93.56±0.49		**93.66±0.29**		89.89±0.15		89.65±0.26		93.92±0.11		**93.95±0.05**		93.32±0.12		93.31±0.19	
	Worst (40.21%)	AUROC Softmax-based	77.80±0.76	82.15±2.72	74.65±0.25	78.14±1.36	85.68±0.09	88.52±0.72	87.57±1.28	86.81±1.99	88.00±1.85	84.97±5.01	88.26±0.45	91.12±0.60	90.55±0.40	93.03±0.18	82.08±0.17	86.02±0.69	77.12±7.08	81.76±5.31	77.35±4.71	79.99±5.25	80.17±1.09	81.54±1.20	84.04±0.21	83.06±1.20	83.76±0.67	82.88±1.24	83.59±0.21	83.51±0.73	83.67±0.37	83.45±0.64
		AUROC Logit-based	75.09±1.07	80.92±1.81	71.06±1.55	72.90±3.19	82.88±0.37	84.34±0.84	82.86±2.25	83.69±4.19	86.58±2.30	85.72±6.40	85.37±1.05	90.29±1.40	89.60±0.76	94.33±0.32	76.21±1.18	83.39±0.44	65.07±2.46	74.24±1.24	74.81±1.65	79.00±2.32	77.54±0.65	81.72±0.63	71.63±5.72	74.69±4.86	75.33±1.72	77.32±0.79	77.14±1.15	83.39±0.79	77.29±1.45	81.92±0.82
		AUROC Distance-based	69.35±0.31	74.92±0.13	73.76±0.09	79.03±0.50	78.89±0.19	83.67±0.47	83.74±1.15	88.19±1.32	83.91±1.32	88.35±2.08	81.69±0.39	87.58±0.63	83.09±0.21	88.77±0.20	73.98±0.82	81.62±1.45	69.87±0.43	78.28±0.38	66.11±1.52	71.26±1.85	70.55±1.27	76.52±1.06	74.72±0.72	81.72±0.99	75.18±0.32	82.11±0.14	73.95±0.08	81.97±0.41	74.08±0.10	81.58±0.28
		AUROC Gradient-based	73.61±1.62	80.76±3.61	34.33±2.77	31.54±4.99	82.34±0.31	83.62±1.13	82.43±3.49	80.77±5.58	88.13±0.58	85.71±5.81	55.96±3.18	70.94±3.29	78.81±1.23	87.08±0.65	71.53±1.99	77.28±1.38	71.49±1.36	78.52±1.36	77.91±0.26	81.67±0.76	73.60±3.89	76.86±5.13	80.38±3.63	83.37±3.43	80.86±0.98	79.39±1.39	79.58±0.84	80.08±2.32	79.70±0.64	79.11±2.52
		Mean	72.72±0.68	78.26±1.13	70.97±0.33	74.45±1.33	**81.40±0.20**	**84.65±0.65**	84.00±1.64	86.16±2.52	85.62±1.44	86.85±3.80	82.53±0.37	88.12±0.60	**86.10±0.29**	**91.11±0.08**	**75.83±0.86**	**82.64±0.93**	69.58±1.38	77.57±0.77	71.25±1.94	75.63±2.40	74.44±0.46	78.97±0.20	75.51±2.32	79.80±2.06	76.88±0.64	80.57±0.19	76.77±0.42	**82.56±0.06**	**76.90±0.38**	81.85±0.14
		ID Accuracy	78.31±0.94		**88.86±0.12**		87.52±0.13		95.63±0.23		**95.80±0.08**		93.71±0.17		93.63±0.09		86.24±0.85		**86.43±0.31**		80.35±0.51		79.87±0.78		**87.72±0.31**		88.01±0.16		84.36±0.12		85.17±0.75	
CIFAR-100N	Clean (0%)	AUROC Softmax-based	81.45±0.14	80.09±0.78	77.13±0.18	77.14±0.42	78.31±0.16	78.50±0.44	78.47±0.31	68.97±3.60	79.15±0.18	71.46±2.03	79.76±0.09	72.95±0.28	79.92±0.41	71.83±0.83	73.05±0.17	74.00±2.22	75.59±0.19	75.92±1.72	78.99±1.25	76.36±2.38	80.92±0.21	77.31±1.22	75.78±0.27	68.51±0.73	75.41±0.57	67.18±0.89	76.95±0.71	70.25±1.87	77.28±0.20	69.78±1.43
		AUROC Logit-based	77.99±0.09	76.22±0.89	75.71±0.45	80.66±0.35	76.56±0.10	80.81±0.62	61.02±1.40	57.19±5.05	61.24±0.95	59.21±2.56	77.07±0.41	74.58±1.38	77.06±0.83	74.94±1.13	50.32±0.44	59.94±0.99	47.15±0.47	58.36±0.62	69.62±1.47	69.69±2.90	68.57±0.52	67.15±0.32	61.44±1.42	62.96±2.31	61.76±1.78	62.41±4.75	64.53±0.95	67.33±2.29	61.23±2.49	62.71±2.14
		AUROC Distance-based	70.41±0.09	76.58±0.43	67.41±0.20	74.71±0.18	68.87±0.12	76.10±0.84	62.39±0.19	65.27±0.57	61.51±0.30	66.34±1.64	69.22±0.19	74.79±0.36	69.42±0.12	74.27±0.13	63.02±0.41	70.47±0.72	61.20±0.09	69.24±0.63	68.15±1.91	75.91±2.92	69.49±0.01	78.24±0.86	66.88±0.14	73.65±0.31	67.12±0.08	72.72±0.81	65.83±0.83	72.72±1.07	65.20±0.44	72.23±0.64
		AUROC Gradient-based	78.10±0.21	75.93±1.53	67.18±0.47	69.27±1.79	66.67±0.98	70.46±2.91	40.11±1.07	38.91±6.26	39.90±1.99	38.92±4.23	65.55±0.45	57.62±2.77	66.43±0.41	58.84±2.47	34.28±0.36	47.66±1.73	37.70±0.45	50.11±1.55	66.80±3.89	62.08±10.98	69.49±0.01	78.24±0.86	70.58±0.84	69.59±4.10	69.55±0.49	73.13±2.13	72.91±1.10	72.53±2.35	72.62±0.57	74.24±0.76
		Mean	74.95±0.05	76.99±0.58	71.55±0.27	76.69±0.25	**72.67±0.13**	**77.67±0.22**	63.32±0.60	61.92±2.47	63.07±0.43	63.45±2.05	73.17±0.13	**73.53±0.59**	**73.33±0.31**	73.28±0.67	59.08±0.30	66.50±0.71	57.80±0.24	65.85±0.37	70.25±0.46	73.29±1.41	**70.79±0.21**	**73.43±0.57**	66.76±0.49	69.25±0.82	66.86±0.57	68.61±2.08	**67.55±0.24**	**70.62±0.94**	66.24±1.04	68.94±0.39
		ID Accuracy	78.85±0.17		75.55±0.14		**75.56±0.11**		77.60±0.23		**77.80±0.22**		73.85±0.17		74.10±0.17		79.76±0.13		**80.31±0.22**		75.90±2.57		77.67±0.22		71.53±0.23		71.55±0.10		71.99±0.59		**72.21±0.29**	
	Noisy (40.20%)	AUROC Softmax-based	71.49±0.36	73.84±0.22	67.44±1.78	68.79±2.79	74.69±0.26	76.17±1.00	76.66±0.83	73.66±2.29	77.25±0.18	75.71±0.64	77.65±0.11	71.37±0.73	77.68±0.12	71.79±1.41	70.74±0.43	70.53±1.23	71.34±0.51	69.60±1.06	73.67±0.07	69.29±0.54	71.42±1.23	66.82±1.36	72.22±0.31	67.17±2.11	72.86±0.72	68.52±0.60	72.70±0.23	67.46±2.00	72.73±0.11	67.09±0.99
		AUROC Logit-based	68.75±0.28	72.55±0.63	65.35±1.70	70.20±1.72	69.61±0.60	73.62±0.37	60.56±1.87	60.36±4.05	60.89±0.74	63.05±3.03	75.18±0.38	73.80±1.04	75.40±0.17	75.61±1.27	54.90±1.24	62.64±0.41	50.67±1.09	59.88±1.93	69.24±0.25	69.12±0.43	67.90±0.77	66.60±1.39	60.94±0.43	64.85±2.33	62.04±1.48	65.32±1.65	62.64±0.91	66.71±1.86	62.55±2.57	64.23±0.65
		AUROC Distance-based	61.50±0.20	70.23±0.60	60.41±1.66	70.79±1.44	63.43±0.32	70.25±0.38	61.38±1.14	65.97±1.31	60.79±0.21	67.86±1.09	66.94±0.22	72.63±0.41	66.93±0.15	73.47±0.75	60.45±0.48	69.44±0.53	60.67±0.05	69.71±0.97	62.15±0.50	70.85±1.20	60.38±0.45	70.65±0.78	64.85±0.05	72.73±0.46	64.93±0.38	73.18±0.98	62.11±0.25	70.61±0.43	62.05±0.20	69.71±0.48
		AUROC Gradient-based	68.40±0.72	70.05±0.25	56.60±0.72	62.26±2.17	43.70±0.75	40.99±1.60	42.28±2.22	40.47±4.35	44.05±0.48	46.44±6.02	64.24±0.15	60.26±3.43	65.03±0.82	63.37±1.27	46.17±2.73	56.40±2.64	45.57±1.12	53.98±1.61	70.70±0.21	67.31±2.05	71.10±0.33	66.32±2.74	69.17±0.92	70.62±1.75	68.22±0.60	70.10±1.08	69.57±0.98	72.85±1.95	69.56±0.35	71.15±4.57
		Mean	65.73±0.12	71.52±0.50	62.88±1.64	69.84±1.52	**66.12±0.41**	**70.71±0.36**	62.53±1.35	64.07±2.44	62.54±0.17	66.45±1.82	71.09±0.24	72.15±0.81	**71.20±0.13**	**73.35±0.96**	59.57±0.38	66.78±0.22	58.40±0.47	65.76±0.87	**66.66±0.32**	**69.87±0.77**	65.06±0.58	68.54±1.13	65.00±0.06	69.25±1.12	**65.44±0.45**	**69.80±0.98**	64.34±0.26	69.00±0.55	64.29±0.91	67.64±0.13
		ID Accuracy	53.91±0.46		**63.96±1.73**		62.50±0.15		70.96±0.18		**70.97±0.21**		66.90±0.05		66.91±0.17		65.29±1.81		**66.90±0.06**		60.44±0.15		60.25±0.31		66.29±0.24		**66.85±0.35**		62.64±0.23		62.99±0.38	

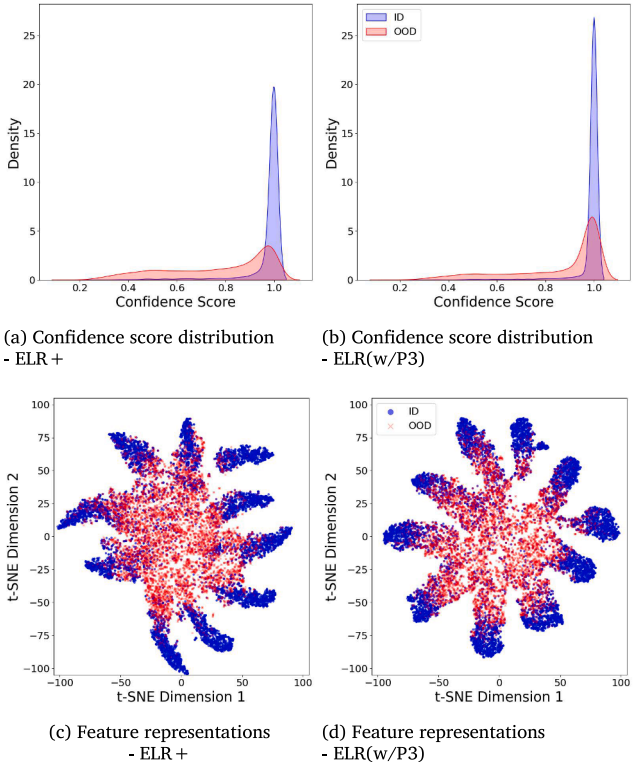
Table 5
Effects of Propositions 2 and 3 on OOD detection performance in terms of AUPR across NLL methods (ELR, ProMix, PGDF, and TCL). The best results are highlighted in **bold**, and the second-best results are underlined.

Dataset	Type	Metric	Baseline (w/o NLL)	ELR				ProMix								PGDF								TCL							
				ELR+		w/P3		Original		w/P2		w/P3		w/P2&P3		Original		w/P2		w/P3		w/P2&P3		Original		w/P2		w/P3		w/P2&P3	
			Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD
CIFAR-10N	Clean (0%)	Softmax-based	85.55±0.71	73.49±2.06	83.19±0.73	70.83±2.97	86.94±0.76	74.38±2.37	83.42±1.15	58.29±2.08	84.88±1.24	64.76±1.23	89.10±0.64	83.27±1.65	92.85±0.40	89.42±1.02	64.39±5.07	49.58±6.14	72.06±6.45	59.10±6.49	88.67±0.15	77.12±0.56	92.79±0.34	87.65±0.21	90.19±0.26	73.96±0.27	91.10±0.15	76.55±1.72	91.39±0.18	80.97±1.29	91.42±0.27	79.20±1.88
		Logit-based	83.86±0.73	71.60±3.21	83.47±0.61	70.73±3.10	86.41±0.62	73.72±1.51	77.26±0.60	54.61±1.44	78.13±2.92	58.69±2.90	85.56±0.42	81.56±1.09	92.07±0.44	90.93±0.85	66.06±6.18	52.59±8.92	55.14±0.90	47.01±1.08	83.49±1.45	73.40±1.35	90.51±0.51	87.51±0.66	81.28±0.71	67.87±2.30	84.82±0.61	75.20±1.34	83.31±1.17	78.64±1.83	86.26±1.11	80.63±2.64
		Distance-based	81.50±0.49	74.25±1.15	77.76±0.50	68.64±2.04	80.12±0.35	70.91±1.06	82.63±0.73	74.16±0.70	82.33±0.42	76.48±1.17	81.67±0.21	78.02±0.71	84.15±0.27	81.21±0.64	75.89±2.89	69.87±4.62	71.17±0.52	66.18±0.12	82.25±0.26	75.01±0.40	84.51±0.32	78.85±0.92	79.68±0.17	73.04±0.27	80.30±0.18	76.01±1.05	78.21±0.17	74.15±0.50	78.62±0.09	73.72±0.14
		Gradient-based	59.66±0.44	37.61±1.23	80.02±0.57	67.81±3.09	82.90±2.36	71.75±4.22	77.43±1.51	51.09±0.28	79.17±4.77	57.54±5.29	53.08±1.63	48.84±5.68	82.17±2.00	80.83±2.69	48.56±7.49	38.23±9.83	49.91±0.82	43.06±1.47	55.73±0.27	42.43±2.63	77.94±0.79	73.12±0.39	88.62±0.87	78.35±1.25	86.66±1.85	79.61±0.89	86.12±2.43	77.07±1.85	87.00±1.08	79.29±3.08
		Mean	81.73±0.46	71.36±1.86	80.54±0.45	69.60±2.13	**83.33±0.59**	**72.39±1.53**	80.78±0.56	64.27±0.60	81.24±1.27	68.02±0.57	82.57±0.27	78.43±0.98	**87.92±0.35**	**85.55±0.74**	69.53±3.65	59.54±5.92	65.13±0.52	57.79±0.67	82.26±0.60	73.12±0.77	**87.37±0.17**	**82.67±0.45**	82.32±0.22	71.83±0.90	**83.77±0.07**	76.03±1.03	82.32±0.32	76.80±0.79	83.49±0.37	**77.06±0.69**
		ID Accuracy	95.62±0.01		**94.15±0.08**		93.68±0.22		96.05±0.01		**96.99±0.08**		94.18±0.15		94.24±0.07		95.68±0.15		**95.97±0.10**		95.46±0.21		95.54±0.15		**94.39±0.16**		94.31±0.23		91.83±0.88		91.72±0.82	
	Aggregate (9.03%)	Softmax-based	82.00±0.53	71.56±2.22	65.57±3.02	46.05±4.26	85.90±0.22	73.31±1.65	83.26±0.49	61.03±4.66	85.54±1.05	65.43±3.87	88.81±0.36	81.00±0.50	92.56±0.15	89.32±0.16	73.79±8.25	61.31±6.62	77.99±1.42	65.08±1.20	84.59±1.08	65.16±1.39	87.27±0.88	73.63±3.51	89.24±0.77	71.85±3.92	89.36±0.34	71.53±2.16	91.82±0.16	80.68±0.64	91.82±0.43	81.73±0.64
		Logit-based	79.24±0.10	63.71±1.59	69.72±1.75	52.16±1.60	81.65±0.31	67.60±0.23	77.14±1.51	56.26±5.65	79.56±3.48	60.85±4.06	85.71±0.21	80.39±0.51	91.32±0.20	90.42±0.45	70.38±2.14	61.33±1.65	59.39±1.08	52.43±1.50	81.33±1.53	63.72±1.60	83.96±0.91	71.83±4.81	82.94±0.23	70.83±2.17	82.66±1.88	70.46±0.49	85.54±0.81	80.58±2.52	88.13±0.96	83.78±0.94
		Distance-based	76.28±0.44	67.18±0.50	73.12±0.70	61.88±0.52	78.30±0.48	70.10±1.78	83.19±0.12	77.65±2.13	80.88±1.21	74.20±2.31	82.21±0.14	77.19±0.46	83.75±0.34	80.87±0.24	76.70±0.66	68.48±0.79	72.46±0.63	65.42±0.57	78.09±0.49	65.83±0.68	78.55±0.72	67.68±1.86	79.90±0.31	74.37±1.22	79.60±0.26	74.15±0.10	78.56±0.29	74.26±0.35	79.43±0.50	74.77±0.39
		Gradient-based	62.38±0.45	38.05±2.63	66.13±3.08	50.93±4.76	74.27±0.68	59.27±0.91	76.94±2.20	55.31±4.23	80.15±3.40	59.68±6.56	54.53±0.53	49.48±1.10	79.31±3.31	79.19±3.26	53.87±2.36	49.05±2.09	54.85±0.82	49.54±1.53	59.92±2.06	43.20±2.63	74.35±2.28	62.97±9.27	88.26±1.69	77.70±5.07	87.94±1.38	77.41±3.19	87.31±1.86	79.25±2.11	87.94±1.17	80.14±1.72
		Mean	77.39±0.21	65.24±1.14	70.49±1.41	55.73±1.36	**80.35±0.21**	**69.25±0.69**	80.96±0.47	67.10±3.45	81.16±2.00	67.84±3.08	82.90±0.10	77.35±0.47	**87.30±0.39**	**85.13±0.25**	73.04±2.26	64.07±1.20	68.28±0.17	60.43±0.81	79.18±0.95	63.87±0.69	**81.41±0.83**	**69.68±3.33**	82.77±0.38	73.03±2.06	82.55±0.53	72.74±0.09	83.32±0.13	77.53±1.01	**84.58±0.54**	**79.00±0.37**
		ID Accuracy	90.48±0.06		**93.27±0.02**		92.41±0.11		96.46±0.48		**96.83±0.05**		94.06±0.16		94.24±0.03		94.06±0.05		**94.14±0.22**		91.71±0.23		91.51±0.24		**94.71±0.28**		94.68±0.02		92.70±0.59		92.61±0.66	
	Random1 (17.23%)	Softmax-based	77.67±3.81	62.02±5.59	64.85±1.42	45.93±2.26	87.73±0.17	75.78±1.43	83.52±1.36	59.69±2.35	84.38±0.74	59.59±3.35	89.30±0.42	83.40±0.68	92.82±0.25	89.30±0.78	66.64±0.96	53.07±2.50	81.94±0.70	68.37±2.00	85.07±0.87	70.55±1.30	86.66±0.38	72.75±3.05	87.41±1.04	67.43±1.66	87.60±0.48	67.80±2.00	91.45±0.23	79.43±0.58	91.56±0.26	81.28±0.87
		Logit-based	76.59±1.60	61.14±3.77	68.72±1.43	52.30±1.93	85.34±0.26	74.42±0.87	78.31±2.70	56.63±2.63	77.62±3.20	54.88±4.77	85.65±0.39	81.82±0.38	91.97±0.24	90.75±0.70	72.32±1.51	60.90±3.16	60.47±2.80	54.60±1.19	81.02±1.12	69.33±1.47	82.63±0.94	71.42±3.23	77.88±1.75	63.32±2.29	84.34±2.77	71.84±3.33	85.36±1.22	76.85±3.88	85.50±0.68	76.96±1.43
		Distance-based	71.74±0.91	60.68±2.55	73.97±0.74	62.52±1.36	80.64±0.54	71.01±0.06	82.42±0.14	75.49±1.21	81.72±1.23	74.54±2.44	81.90±0.50	78.08±0.05	83.97±0.04	80.74±0.21	76.30±0.42	68.84±1.53	72.63±0.25	65.24±1.73	76.47±0.41	66.14±0.54	76.79±0.25	65.17±1.85	78.56±0.37	73.47±0.60	79.10±0.32	73.08±0.83	78.90±0.43	73.74±1.03	79.25±0.21	74.66±0.32
		Gradient-based	72.18±3.19	54.93±4.74	55.93±1.43	37.84±1.78	82.38±0.22	72.95±0.70	74.18±2.69	46.93±2.02	80.29±5.46	55.62±8.48	55.17±0.76	50.99±2.56	81.68±1.94	79.63±3.38	60.34±2.43	49.88±3.71	60.07±1.65	54.28±1.38	61.93±0.46	47.55±3.58	71.45±2.24	56.65±2.74	85.78±1.59	74.82±2.18	86.42±1.23	76.68±3.32	87.67±0.68	78.96±1.73	86.22±0.83	75.43±2.39
		Mean	74.23±0.44	60.73±2.68	69.93±0.93	55.37±1.48	**83.33±0.34**	**72.94±0.49**	80.86±1.06	65.54±0.64	80.77±1.99	64.97±3.50	82.84±0.27	78.68±0.29	**87.77±0.20**	**85.20±0.58**	72.68±0.84	62.84±2.24	69.60±1.09	61.80±1.29	78.50±0.71	66.86±1.06	**79.91±0.46**	**67.89±2.48**	80.12±0.66	69.39±0.50	82.48±0.99	72.05±0.72	83.38±0.64	75.89±1.87	**83.53±0.31**	**76.47±0.64**
		ID Accuracy	84.86±0.29		**92.74±0.05**		92.23±0.17		96.37±0.22		96.48±0.13		94.00±0.05		**94.26±0.09**		93.56±0.49		**93.66±0.29**		89.89±0.15		89.65±0.26		93.92±0.11		**93.95±0.05**		93.32±0.12		93.31±0.19	
	Worst (40.21%)	Softmax-based	79.37±0.70	66.94±3.09	66.08±0.38	47.15±1.73	87.43±0.17	78.05±0.87	86.80±1.35	70.49±4.49	87.89±2.02	69.53±8.92	88.51±1.10	82.68±2.54	92.13±0.57	88.28±0.47	82.36±0.83	68.53±1.16	77.74±6.27	64.16±5.77	78.84±5.10	63.03±7.76	82.12±0.90	65.55±1.23	85.84±0.16	65.73±1.07	85.46±0.75	65.69±1.66	86.12±0.27	69.73±1.19	86.30±0.54	69.70±1.16
		Logit-based	76.58±1.05	66.28±1.66	66.20±1.21	45.53±3.45	84.62±0.37	71.63±1.04	81.02±2.37	64.92±7.29	85.00±3.05	69.91±9.48	84.40±1.36	79.41±3.20	90.50±0.92	89.28±0.45	75.21±1.58	64.92±0.34	64.82±2.69	55.02±1.62	76.47±1.44	61.67±3.15	78.90±0.36	65.10±1.69	74.85±4.73	60.57±4.26	78.75±1.48	63.12±1.22	80.15±1.05	71.31±0.56	80.37±1.33	69.69±0.16
		Distance-based	70.66±0.35	60.18±0.40	72.35±0.11	62.08±1.35	79.14±0.15	70.43±0.58	83.24±1.21	77.25±2.15	82.78±1.25	76.59±2.45	81.11±0.25	76.08±1.19	83.36±0.28	79.29±0.14	74.46±0.96	68.06±2.35	70.61±0.59	63.50±0.55	68.11±1.32	55.70±1.68	71.43±1.09	59.93±1.28	75.71±0.68	68.98±1.38	76.12±0.31	69.45±0.65	74.97±0.17	69.24±0.73	75.07±0.13	68.61±0.72
		Gradient-based	74.27±0.96	63.28±3.66	40.35±1.03	22.23±1.07	81.42±0.43	66.16±0.44	79.80±3.73	59.46±9.09	86.21±1.07	69.63±8.26	54.81±2.69	50.29±5.10	76.79±1.67	75.35±0.99	68.92±2.27	57.29±1.95	68.68±2.57	57.59±1.57	75.95±0.80	58.06±1.57	71.41±4.23	53.23±7.74	83.72±2.97	74.04±4.14	84.10±0.94	66.65±3.55	82.85±0.52	68.89±4.27	82.93±0.74	67.43±3.01
		Mean	74.09±0.63	63.34±1.19	67.73±0.27	52.40±1.87	**82.30±0.18**	**71.78±0.71**	82.92±1.73	71.35±4.42	84.47±1.74	73.00±5.60	81.93±0.68	76.82±1.93	**86.65±0.53**	**83.66±0.21**	**75.65±1.17**	**66.57±1.48**	69.80±1.70	60.62±1.41	72.86±1.82	58.87±3.05	75.48±0.25	62.10±1.00	77.46±1.96	66.08±2.11	78.85±0.55	66.71±1.00	78.78±0.36	**69.95±0.18**	**78.93±0.31**	69.06±0.64
		ID Accuracy	78.31±0.94		**88.86±0.12**		87.52±0.13		95.63±0.23		**95.80±0.08**		93.71±0.17		93.63±0.09		86.24±0.85		**86.43±0.31**		80.35±0.51		79.87±0.78		87.72±0.31		**88.01±0.16**		84.36±0.12		85.17±0.75	
CIFAR-100N	Clean (0%)	Softmax-based	84.30±0.09	68.49±1.44	80.74±0.22	66.94±0.52	81.48±0.21	67.85±1.32	79.62±0.27	53.91±2.64	80.42±0.24	55.58±1.92	83.75±0.06	62.01±1.03	83.83±0.40	60.27±1.65	74.41±0.59	55.37±2.50	77.04±0.09	58.14±2.14	81.45±1.13	63.95±3.56	83.84±0.44	65.22±2.37	78.64±0.47	52.05±0.11	78.43±0.91	51.20±0.64	80.18±1.22	55.68±2.64	80.85±0.27	55.17±1.66
		Logit-based	80.90±0.07	63.84±1.55	78.38±0.27	70.90±0.71	79.39±0.15	70.37±1.31	64.26±1.28	44.71±4.67	64.01±0.67	46.10±1.82	81.04±0.43	63.01±1.91	81.10±0.82	63.03±1.54	56.53±0.48	44.18±1.44	54.47±0.27	44.37±0.19	72.45±1.12	57.06±3.71	72.13±0.71	54.29±0.81	66.46±1.43	47.77±2.90	66.58±1.49	48.67±4.12	68.90±0.36	52.62±2.49	65.48±1.75	48.79±2.78
		Distance-based	73.86±0.07	64.03±0.48	70.78±0.22	62.04±0.54	72.09±0.23	63.14±0.87	66.76±0.21	52.41±0.48	65.99±0.11	54.06±2.05	72.70±0.22	62.93±0.63	72.86±0.24	62.26±0.58	66.42±0.48	55.67±1.46	64.87±0.10	54.30±1.02	70.19±1.41	62.05±3.13	71.69±0.48	64.69±1.58	70.44±0.19	61.03±0.66	70.43±0.14	60.23±1.17	69.54±0.89	60.00±1.41	68.72±0.41	59.22±0.57
		Gradient-based	77.69±0.30	54.01±1.62	71.40±0.47	54.52±3.03	70.75±1.07	56.22±4.21	47.82±0.55	28.26±2.81	47.66±1.29	28.12±2.00	66.54±0.08	39.79±2.32	67.69±0.16	40.65±1.68	44.09±0.25	31.66±0.33	46.57±0.20	35.99±0.36	66.98±4.48	43.82±12.31	65.16±0.34	35.19±2.21	76.91±0.93	61.36±4.26	75.68±0.76	64.93±2.47	78.56±0.95	63.91±1.97	78.13±0.69	65.02±2.18
		Mean	**77.93±0.04**	64.15±0.85	74.79±0.23	65.21±0.06	**75.81±0.22**	**65.80±0.73**	67.00±0.56	48.94±2.17	66.68±0.27	50.42±1.92	76.75±0.11	**61.59±1.03**	**76.92±0.27**	61.05±1.09	63.38±0.44	50.73±1.45	62.55±0.06	50.80±0.56	72.51±0.28	59.81±2.02	**73.41±0.51**	**59.93±1.36**	70.82±0.54	55.44±1.26	70.75±0.50	55.40±2.01	**71.49±0.36**	**57.19±0.81**	70.11±0.79	55.59±0.52
		ID Accuracy	78.85±0.17		75.55±0.14		**75.56±0.11**		77.60±0.23		**77.80±0.22**		73.85±0.17		74.10±0.17		79.76±0.13		**80.31±0.22**		75.90±2.57		77.67±0.22		71.53±0.23		71.55±0.10		71.99±0.59		**72.21±0.29**	
	Noisy (40.20%)	Softmax-based	74.43±0.28	60.10±0.78	71.16±1.51	55.90±3.06	76.23±0.67	64.43±1.51	78.56±0.99	56.24±3.44	79.48±0.45	58.23±1.55	81.62±0.21	59.12±0.82	81.71±0.13	59.59±2.42	73.13±0.48	52.60±0.53	73.63±1.05	50.32±1.01	77.22±0.05	54.60±1.29	76.07±0.20	54.10±3.10	74.00±0.77	49.04±1.10	74.71±1.17	49.65±0.66	74.52±0.85	50.35±2.29	74.81±0.22	50.15±1.55
		Logit-based	71.72±0.27	58.02±1.33	68.54±1.54	56.29±1.90	71.45±0.83	60.62±0.77	64.34±1.82	45.96±3.43	64.04±0.74	47.16±3.09	79.09±0.48	60.81±1.25	79.53±0.11	61.82±1.86	59.79±1.03	46.10±1.39	56.33±1.01	42.80±1.72	73.00±0.18	53.57±1.35	71.83±0.58	52.26±2.34	65.22±0.16	47.84±1.67	66.40±1.81	48.34±1.58	66.31±0.91	50.87±1.79	66.20±2.71	48.93±0.31
		Distance-based	65.46±0.15	55.76±0.57	63.85±1.77	55.75±1.21	66.25±0.29	55.51±0.98	65.95±0.91	52.08±1.51	65.12±0.25	54.65±0.54	70.43±0.16	59.46±0.45	70.56±0.25	59.92±1.06	63.84±0.35	53.81±0.77	64.17±0.13	54.05±0.72	65.76±0.44	55.30±1.56	64.29±0.36	54.65±1.14	68.48±0.16	59.05±0.71	68.72±0.37	60.33±1.29	65.81±0.22	55.92±0.91	65.67±0.31	55.25±0.40
		Gradient-based	69.43±0.71	48.71±0.88	59.10±0.46	45.03±4.06	48.26±0.60	26.76±0.67	48.47±1.20	28.45±1.41	49.52±0.44	32.25±4.06	65.62±0.32	41.44±3.37	67.01±0.54	44.56±1.12	51.78±2.36	39.10±3.48	51.09±0.92	36.99±1.62	73.58±0.43	47.73±3.59	74.92±0.33	50.40±2.44	74.74±1.38	61.13±1.14	74.40±0.33	60.83±0.51	74.24±1.18	61.99±3.88	74.16±0.41	60.59±5.76
		Mean	69.06±0.12	56.79±0.82	66.23±1.57	55.38±1.47	**68.52±0.52**	**57.02±0.91**	66.51±1.20	49.56±2.40	66.22±0.25	51.67±1.65	74.68±0.26	58.88±0.87	**74.96±0.18**	**59.66±1.46**	63.39±0.35	50.41±0.73	62.50±0.58	49.01±0.65	**70.27±0.25**	**54.24±1.29**	69.09±0.33	53.58±1.84	68.65±0.21	54.04±0.96	**69.23±0.61**	**54.88±1.08**	67.78±0.44	53.77±0.70	67.73±1.04	52.73±0.11
		ID Accuracy	53.91±0.46		**63.96±1.73**		62.50±0.15		70.96±0.18		**70.97±0.21**		66.90±0.05		66.91±0.17		65.29±1.81		**66.90±0.06**		60.44±0.15		60.25±0.31		66.29±0.24		**66.85±0.35**		62.64±0.23		62.99±0.38	



(a) Confidence score distribution - Baseline on *Clean* (0%)
(b) Confidence score distribution - Baseline on *Random1* (17.23%)
(c) Confidence score distribution - SOP on *Random1* (17.23%)
(d) Confidence score distribution - ProMix on *Random1* (17.23%)
(e) Feature representations - Baseline on *Clean* (0%)
(f) Feature representations - Baseline on *Random1* (17.23%)
(g) Feature representations - SOP on *Random1* (17.23%)
(h) Feature representations - ProMix on *Random1* (17.23%)

Fig. 5. Comparison of ID–OOD separability between the baseline (w/o NLL) and the use of NLL methods (SOP, ProMix) trained on CIFAR-10N with *Random1* (17.23%). For visualization, ID instances are drawn from CIFAR-10, and OOD instances from CIFAR-100. (a) and (e) correspond to the clean label baseline. (a–d) show the softmax confidence distributions. (e–h) show the t-SNE visualizations of feature representations extracted from the penultimate layer.



(a) Confidence score distribution - ELR+
(b) Confidence score distribution - ELR(w/P3)
(c) Feature representations - ELR+
(d) Feature representations - ELR(w/P3)

Fig. 6. Effect of Proposition 3 on ID-OOD separability of ELR trained on CIFAR-10N with *Random1* (17.23%). For visualization, ID instances are drawn from CIFAR-10, and OOD instances from CIFAR-100. (a–b) show the softmax confidence distributions. (c-d) show the t-SNE visualizations of feature representations extracted from the penultimate layer.

advantageous for OOD detection and can even interfere with the separation between ID and OOD instances.

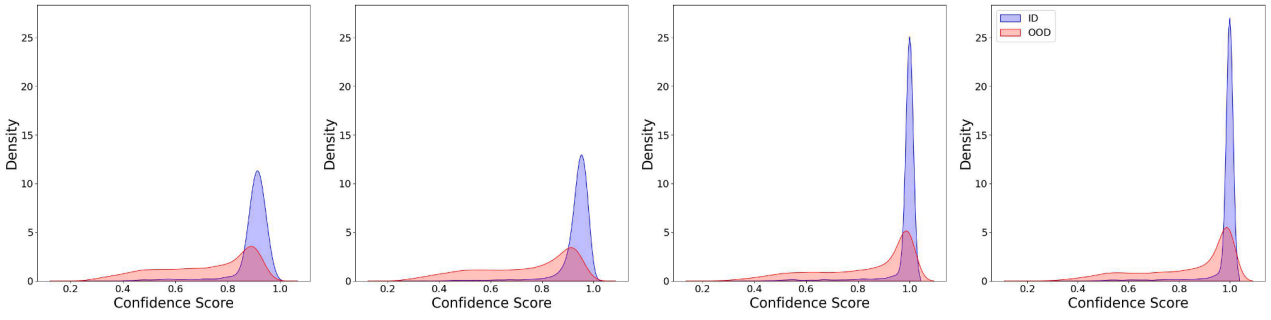
For SSL-based NLL methods (ProMix, PGDF, and TCL), introducing both Propositions 2 and 3 consistently yielded better OOD detection performance than applying either proposition individually, suggesting a complementary effect between the two propositions. Proposition 2 drives higher confidence on clean ID instances by explicitly minimizing their energy. Proposition 3 mitigates the confidence ceiling by eliminating Mixup-related components from NLL. Together, they yield a synergistic improvement in OOD detection.

6.4. Additional analyses

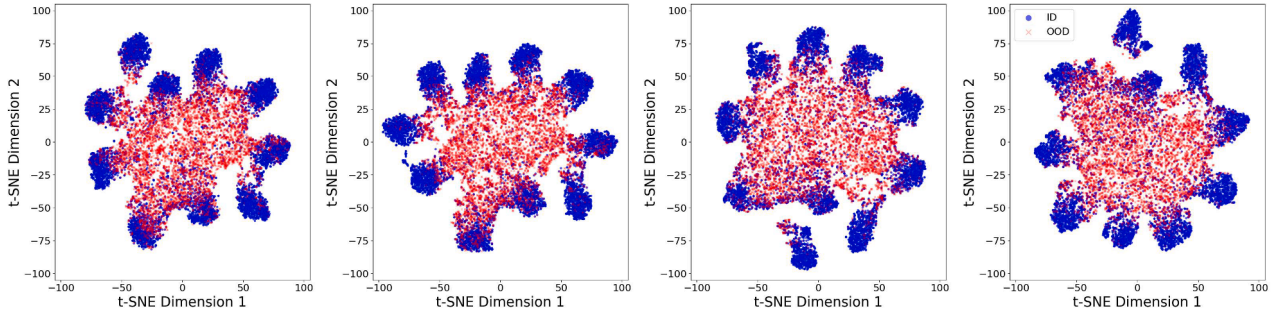
6.4.1. Effects of Propositions 2 and 3 on baseline performance

We investigated the effects of Propositions 2 and 3 on the performance of the baseline that trains a model without NLL. For Proposition 2, we examined whether model output sharpening through energy minimization improves ID classification and OOD detection performance. When applying model output sharpening (*i.e.*, w/P2), the labeled set $\mathcal{D}_L$ consisted of instances that were correctly classified by the model with confidence greater than 0.7 at each training iteration. For Proposition 3, we examined the effect of Mixup on OOD detection performance. When applying Mixup (*i.e.*, w/o P3), the model was trained using Mixup augmented instances with a mixing coefficient $\lambda \sim \text{Beta}(0.2, 0.2)$.

Tables 6 and 7 present the results in terms of AUROC and AUPR. Integrating model output sharpening into the baseline generally improved both ID classification and OOD detection performance, whereas introducing Mixup consistently and substantially degraded OOD detection performance despite improving ID classification accuracy. These findings are consistent with Propositions 2 and 3, supporting both propositions even when NLL is not applied.

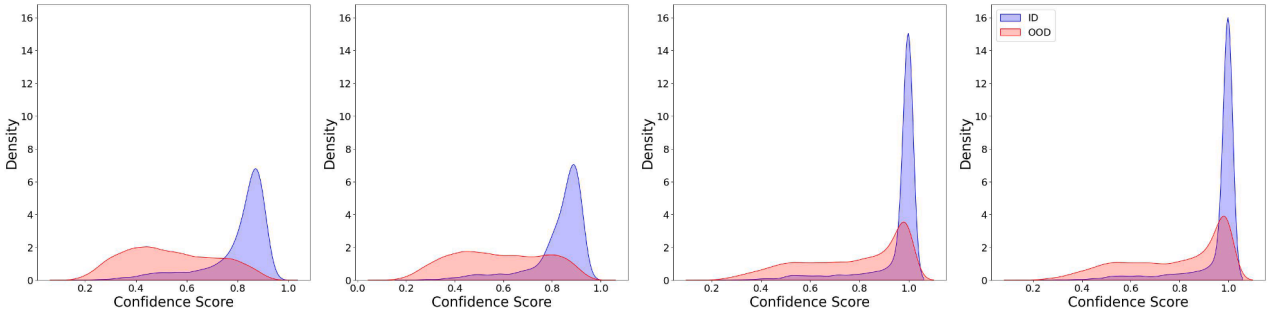


(a) Confidence score distribution - ProMix(Original)
(b) Confidence score distribution - ProMix(w/P2)
(c) Confidence score distribution - ProMix(w/P3)
(d) Confidence score distribution - ProMix(w/P2&P3)

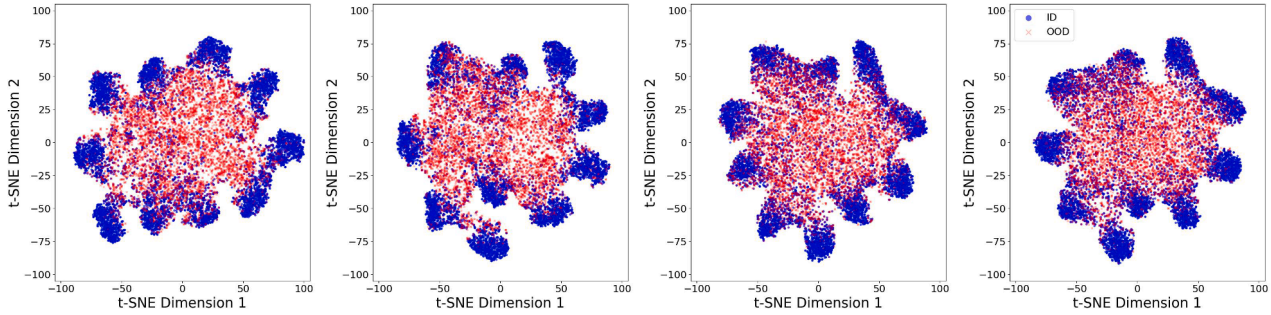


(e) Feature representations - ProMix(Original)
(f) Feature representations - ProMix(w/P2)
(g) Feature representations - ProMix(w/P3)
(h) Feature representations - ProMix(w/P2&P3)

Fig. 7. Effects of Propositions 2 and 3 on ID-OOD separability of ProMix trained on CIFAR-10N with *Random1* (17.23%). For visualization, ID instances are drawn from CIFAR-10, and OOD instances from CIFAR-100. (a–d) show the softmax confidence distributions. (e-h) show the t-SNE visualizations of feature representations extracted from the penultimate layer.

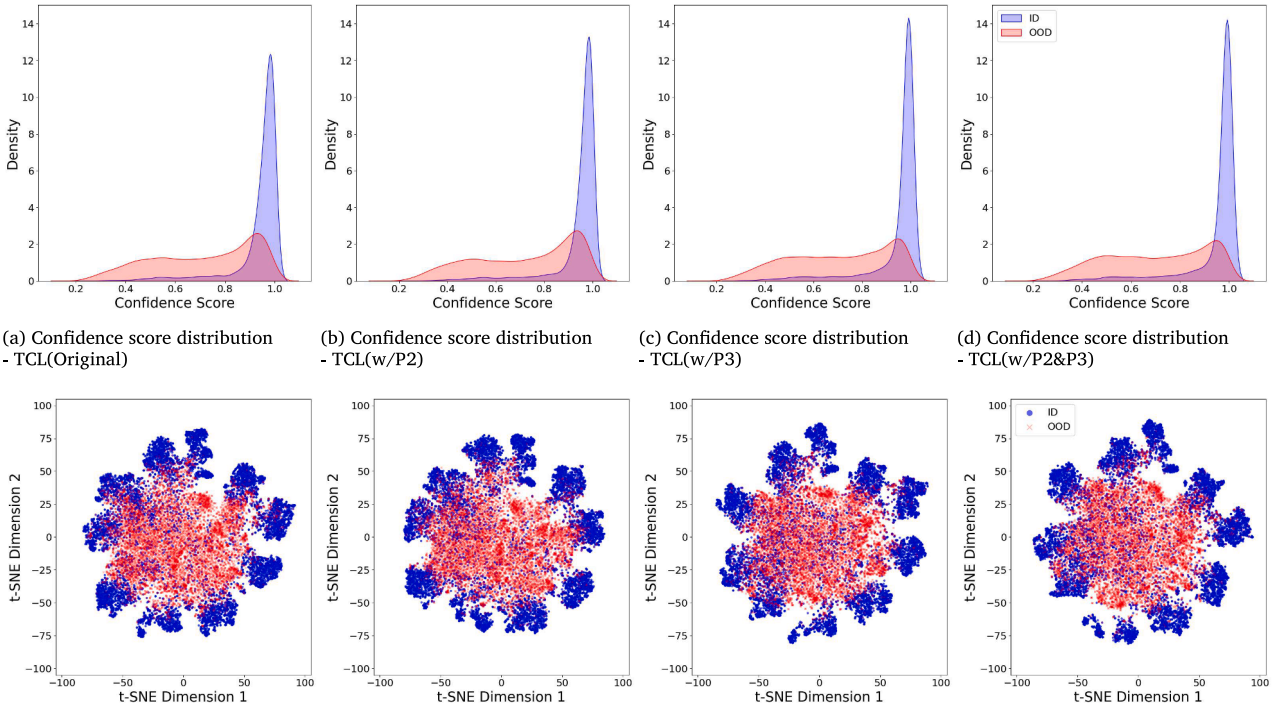


(a) Confidence score distribution - PGDF(Original)
(b) Confidence score distribution - PGDF(w/P2)
(c) Confidence score distribution - PGDF(w/P3)
(d) Confidence score distribution - PGDF(w/P2&P3)



(e) Feature representations - PGDF(Original)
(f) Feature representations - PGDF(w/P2)
(g) Feature representations - PGDF(w/P3)
(h) Feature representations - PGDF(w/P2&P3)

Fig. 8. Effects of Propositions 2 and 3 on ID-OOD separability of PGDF trained on CIFAR-10N with *Random1* (17.23%). For visualization, ID instances are drawn from CIFAR-10, and OOD instances from CIFAR-100. (a–d) show the softmax confidence distributions. (e-h) show the t-SNE visualizations of feature representations extracted from the penultimate layer.



(a) Confidence score distribution - TCL(Original)
(b) Confidence score distribution - TCL(w/P2)
(c) Confidence score distribution - TCL(w/P3)
(d) Confidence score distribution - TCL(w/P2&P3)
(e) Feature representations - TCL(Original)
(f) Feature representations - TCL(w/P2)
(g) Feature representations - TCL(w/P3)
(h) Feature representations - TCL(w/P2&P3)

Fig. 9. Effects of Propositions 2 and 3 on ID-OOD separability of TCL trained on CIFAR-10N with *Random1* (17.23%). For visualization, ID instances are drawn from CIFAR-10, and OOD instances from CIFAR-100. (a–d) show the softmax confidence distributions. (e-h) show the t-SNE visualizations of feature representations extracted from the penultimate layer.

6.4.2. Training time comparison

Table 8 compares the training time of the baseline and NLL methods. Training a model with an NLL method generally increased the training time compared with the baseline (w/o NLL), as NLL methods required additional procedures. This computational overhead was particularly noticeable for SSL-based methods. However, Proposition 2 and 3 did not exacerbate this overhead. While applying Proposition 2 incurred no substantial increase in training time, applying Proposition 3 substantially reduced the training time of ELR, ProMix and TCL. Furthermore, combining both propositions yielded nearly identical training times to that of Proposition 3 alone for ProMix, PGDF and TCL.

Table 6
Effects of Propositions 2 and 3 on baseline performance in terms of AUROC under various noisy label scenarios on CIFAR-10N and CIFAR-100N. The best results are highlighted in **bold**, and the second-best results are underlined.

Dataset	Type	Metric		Baseline		Baseline + Sharpening		Baseline + Mixup		Baseline + Both	
				Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD
CIFAR-10N	Clean (0%)	AUROC	Softmax-based	87.68±0.50	90.36±0.50	89.95±0.42	93.14±0.11	78.81±2.53	80.02±2.31	85.31±6.52	86.71±9.03
			Logit-based	85.67±0.77	87.74±1.97	87.45±0.72	89.81±1.06	75.23±1.34	76.15±1.28	82.22±7.46	83.99±8.17
			Distance-based	82.43±0.57	86.93±0.81	84.63±0.80	89.53±0.45	83.04±0.99	87.63±1.10	81.87±1.97	86.52±2.34
			Gradient-based	63.15±0.84	64.84±1.56	72.08±0.95	76.14±1.62	57.11±1.09	53.82±1.35	69.38±3.06	71.02±6.00
			Mean	83.26±0.58	86.56±1.15	**85.70±0.51**	**89.49±0.31**	78.54±0.80	81.02±0.65	81.87±4.47	84.94±5.41
		ID Accuracy		95.62±0.01		95.31±0.18		**96.06±0.14**		95.45±0.31	
	Aggregate (9.03%)	AUROC	Softmax-based	83.24±0.33	87.10±0.41	84.61±0.51	88.50±0.43	77.28±0.66	80.20±1.88	77.77±2.45	78.96±2.58
			Logit-based	79.85±0.12	80.99±1.12	80.45±0.52	82.39±0.60	70.39±1.04	68.62±1.35	67.08±0.99	63.72±2.17
			Distance-based	76.29±0.37	81.52±0.51	77.78±0.99	83.79±1.48	77.77±0.97	82.86±0.71	74.04±1.14	77.04±0.93
			Gradient-based	65.87±0.52	65.05±1.58	70.39±0.26	74.24±1.46	59.78±3.91	51.52±3.47	64.07±4.15	54.50±5.99
			Mean	77.97±0.18	81.37±0.56	**79.31±0.43**	**83.59±0.87**	74.42±0.79	76.30±0.51	71.91±0.69	71.95±1.16
		ID Accuracy		90.48±0.06		90.44±0.14		**91.62±0.28**		91.57±0.39	
	Random1 (17.23%)	AUROC	Softmax-based	76.52±3.84	78.23±5.12	81.58±1.00	86.67±1.47	73.89±6.63	75.41±7.62	79.93±1.92	83.28±3.28
			Logit-based	75.72±1.17	77.30±2.38	75.48±3.17	81.45±4.66	73.03±2.53	72.27±4.83	70.35±5.50	71.26±10.39
			Distance-based	70.88±0.84	75.46±1.11	71.62±1.09	77.38±2.29	75.04±2.92	79.46±2.68	70.32±1.35	74.91±2.99
			Gradient-based	73.43±2.02	74.46±4.65	74.47±4.15	80.30±5.74	66.56±4.02	62.39±10.20	69.99±4.66	68.00±11.95
			Mean	73.44±0.17	76.42±0.99	**74.56±1.88**	**80.29±3.02**	73.77±1.19	75.65±1.33	71.83±2.61	74.71±5.46
		ID Accuracy		84.86±0.29		86.08±0.05		**88.50±2.38**		86.96±0.66	
	Worst (40.21%)	AUROC	Softmax-based	77.80±0.76	82.15±2.72	76.86±1.51	80.99±0.18	78.74±0.68	84.54±0.92	78.66±1.04	84.97±0.70
			Logit-based	75.09±1.07	80.92±1.81	71.05±3.20	76.26±4.30	73.40±2.23	78.49±2.47	69.39±2.99	74.01±4.32
			Distance-based	69.35±0.31	74.92±0.13	67.31±0.65	72.49±0.22	70.74±1.08	76.50±0.81	66.78±0.75	69.87±1.79
			Gradient-based	73.61±1.62	80.76±3.61	71.90±3.38	78.08±3.86	70.40±2.44	77.46±4.35	71.20±2.80	76.51±4.15
			Mean	72.72±0.68	78.26±1.13	70.24±1.65	75.32±1.69	**72.83±1.43**	**78.45±1.15**	69.71±1.61	73.91±2.53
		ID Accuracy		78.31±0.94		78.56±0.97		**80.32±0.41**		80.16±0.60	
CIFAR-100N	Clean (0%)	AUROC	Softmax-based	81.45±0.14	80.09±0.78	81.18±0.32	81.18±1.04	80.53±0.58	79.14±1.29	80.34±0.40	79.73±1.57
			Logit-based	77.99±0.09	76.22±0.89	77.57±0.40	80.26±1.86	72.87±2.10	72.33±2.08	72.33±3.55	74.05±5.68
			Distance-based	70.41±0.09	76.58±0.43	72.01±1.07	77.79±0.55	72.83±0.68	77.29±0.12	68.93±1.04	73.66±2.59
			Gradient-based	78.10±0.21	75.93±1.53	77.23±1.03	76.90±2.84	58.81±2.24	54.57±1.81	63.68±8.73	61.26±11.56
			Mean	74.95±0.05	76.99±0.58	**75.49±0.51**	**79.06±0.86**	73.32±1.16	74.82±0.93	71.53±2.13	74.09±3.85
		ID Accuracy		78.85±0.17		78.47±0.34		**79.51±0.26**		79.04±0.80	
	Noisy (40.20%)	AUROC	Softmax-based	71.49±0.36	73.84±0.22	71.50±0.44	74.34±0.65	67.44±5.70	68.11±7.79	71.34±0.34	72.36±1.10
			Logit-based	68.75±0.28	72.55±0.63	68.35±0.38	73.22±1.18	64.44±0.65	64.66±1.92	62.60±0.59	62.59±0.70
			Distance-based	61.50±0.20	70.23±0.60	62.00±0.91	70.71±0.71	62.80±0.24	71.47±0.19	60.23±0.16	69.43±1.06
			Gradient-based	68.40±0.72	70.05±0.25	68.29±1.10	70.78±1.06	57.54±0.71	54.77±2.37	58.28±1.45	55.02±2.42
			Mean	65.73±0.12	71.52±0.50	**65.83±0.58**	**72.08±0.78**	63.77±1.22	67.91±1.92	62.63±0.26	66.98±0.53
		ID Accuracy		53.91±0.46		54.47±0.14		56.61±0.26		**56.62±0.66**	

Table 7
Effects of Propositions 2 and 3 on baseline performance in terms of AUPR under various noisy label scenarios on CIFAR-10N and CIFAR-100N. The best results are highlighted in **bold**, and the second-best results are underlined.

Dataset	Type	Metric		Baseline		Baseline + Sharpening		Baseline + Mixup		Baseline + Both	
				Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD	Near-OOD	Far-OOD
CIFAR-10N	Clean (0%)	AUPR	Softmax-based	85.55±0.71	73.49±2.06	90.15±0.48	85.35±0.94	73.20±4.32	52.71±3.13	83.86±8.81	72.49±17.58
			Logit-based	83.86±0.73	71.60±3.21	87.36±0.88	79.61±1.24	72.03±1.46	51.75±1.28	81.13±8.82	68.40±15.14
			Distance-based	81.50±0.49	74.25±1.15	84.43±0.64	80.32±0.79	81.32±0.63	75.37±0.71	81.30±2.63	74.82±4.51
			Gradient-based	59.66±0.44	37.61±1.23	69.32±1.29	51.74±2.76	52.87±1.70	29.03±1.04	65.82±4.00	44.57±7.53
			Mean	81.73±0.46	71.36±1.86	**85.47±0.44**	**79.39±0.21**	75.61±1.09	61.89±0.80	80.84±5.63	70.83±10.07
		ID Accuracy		95.62±0.01		95.31±0.18		**96.06±0.14**		95.45±0.31	
	Aggregate (9.03%)	AUPR	Softmax-based	82.00±0.53	71.56±2.22	85.22±0.52	78.28±0.58	72.99±1.08	52.69±3.10	73.78±4.05	50.94±5.76
			Logit-based	79.24±0.10	63.71±1.59	68.62±0.92	45.60±1.48	68.61±0.94	45.61±1.33	65.05±1.96	41.08±2.11
			Distance-based	76.28±0.44	67.18±0.50	78.20±1.07	71.63±2.19	77.43±0.83	69.45±0.60	73.56±0.91	61.04±0.91
			Gradient-based	62.38±0.45	38.05±2.63	67.55±0.69	50.99±1.63	56.25±2.93	29.26±1.53	59.73±4.90	30.90±3.35
			Mean	77.39±0.21	65.24±1.14	**79.65±0.48**	**70.88±1.27**	72.83±0.69	57.16±0.79	70.18±1.24	51.56±1.29
		ID Accuracy		90.48±0.06		90.44±0.14		**91.62±0.28**		91.57±0.39	
	Random1 (17.23%)	AUPR	Softmax-based	77.67±3.81	62.02±5.59	83.14±1.20	74.36±1.38	73.07±5.70	55.95±9.26	79.86±3.88	64.51±8.49
			Logit-based	76.59±1.60	61.14±3.77	77.24±2.85	67.59±4.46	72.05±3.82	53.09±8.11	70.52±6.20	52.99±13.34
			Distance-based	71.74±0.91	60.68±2.55	72.62±1.02	63.18±2.63	75.29±2.38	64.83±3.43	70.91±1.10	59.21±3.27
			Gradient-based	72.18±3.19	54.93±4.74	73.53±4.21	60.55±5.61	62.80±4.56	39.30±7.29	67.23±5.94	46.28±13.70
			Mean	74.23±0.44	60.73±2.68	**75.79±1.75**	**66.20±2.89**	73.26±0.88	58.38±2.47	72.01±3.14	57.40±7.36
		ID Accuracy		84.86±0.29		86.08±0.05		**88.50±2.38**		86.96±0.66	
	Worst (40.21%)	AUPR	Softmax-based	79.37±0.70	66.94±3.09	77.97±1.96	64.10±1.49	79.88±0.39	70.34±2.22	79.95±1.57	73.35±2.15
			Logit-based	76.58±1.05	66.28±1.66	72.86±3.07	59.56±4.37	73.88±2.17	62.29±1.68	70.76±3.03	58.75±5.34
			Distance-based	70.66±0.35	60.18±0.40	68.74±0.83	56.45±0.60	72.78±1.08	63.67±1.17	68.28±1.01	55.08±1.84
			Gradient-based	74.27±0.96	63.28±3.66	72.26±3.70	57.91±6.60	69.38±2.14	57.83±2.98	71.36±2.94	59.40±6.31
			Mean	**74.09±0.63**	63.34±1.19	71.68±1.78	58.72±2.19	74.07±1.36	**63.98±0.58**	71.07±1.84	59.35±3.08
		ID Accuracy		78.31±0.94		78.56±0.97		**80.32±0.41**		80.16±0.60	
CIFAR-100N	Clean (0%)	AUPR	Softmax-based	84.30±0.09	68.49±1.44	84.04±0.21	70.28±2.37	82.92±0.70	66.02±3.08	82.89±0.70	65.97±4.48
			Logit-based	80.90±0.07	63.84±1.55	80.76±0.32	68.44±2.96	75.11±2.13	55.57±3.63	75.19±3.85	57.44±9.78
			Distance-based	73.86±0.07	64.03±0.48	75.10±1.01	65.47±0.67	75.12±0.70	64.11±0.71	71.76±1.41	60.52±3.74
			Gradient-based	77.69±0.30	54.01±1.62	76.87±1.07	56.19±5.65	59.47±1.38	31.93±0.31	63.97±8.20	40.50±12.63
			Mean	77.93±0.04	64.15±0.85	**78.39±0.48**	**66.68±1.81**	75.52±1.15	60.02±1.93	74.19±2.42	59.35±6.17
		ID Accuracy		78.85±0.17		78.47±0.34		**79.51±0.26**		79.04±0.80	
	Noisy (40.20%)	AUPR	Softmax-based	74.43±0.28	60.10±0.78	74.42±0.56	60.73±1.48	70.66±4.90	53.08±7.13	74.09±0.43	56.06±1.72
			Logit-based	71.72±0.27	58.02±1.33	71.44±0.55	59.31±1.87	67.51±0.67	46.58±2.01	65.99±0.72	44.59±0.99
			Distance-based	65.46±0.15	55.76±0.57	65.74±1.02	56.59±0.77	65.98±0.23	56.22±0.31	63.58±0.15	53.61±1.18
			Gradient-based	69.43±0.71	48.71±0.88	69.35±1.35	50.01±1.70	59.51±1.06	32.96±1.83	60.05±1.28	33.28±1.51
			Mean	69.06±0.12	56.79±0.82	**69.10±0.65**	**57.76±1.22**	66.86±1.10	51.46±1.92	65.82±0.37	50.08±0.69
		ID Accuracy		53.91±0.46		54.47±0.14		56.61±0.26		**56.62±0.66**	

Table 8
Comparison of training time across different NLL methods. Models are trained on CIFAR-10N with *Random1* (17.23%).

NLL Method		No. Training Epochs	Training Time (hr)
Baseline (w/o NLL)		300	1.02
SOP	**Original**	300	3.23
ELR	**ELR+**	200	1.73
	w/P3	120	0.66
ProMix	**Original**	300	9.00
	w/P2	300	9.62
	w/P3	300	4.29
	w/P2&P3	300	4.28
PGDF	**Original**	150	6.17
	w/P2	150	6.21
	w/P3	150	6.31
	w/P2&P3	150	6.35
TCL	**Original**	200	5.36
	w/P2	200	5.57
	w/P3	200	3.23
	w/P2&P3	200	3.34

7. Conclusion

In this study, we proposed a novel framework that leverages NLL methods integrated with post-hoc OOD detection methods for robust OOD detection in the presence of noisy labels. To align NLL methods with OOD detection, we introduced three key propositions: (1) NLL enhances OOD detection under noisy labels; (2) Sharpening model outputs in SSL-based NLL benefits OOD detection; and (3) Mixup in NLL can degrade OOD detection despite ID accuracy gains. Comprehensive experiments on CIFAR-10N, CIFAR-100N, and Tiny-ImageNet benchmarks demonstrated the effectiveness of these propositions in enhancing OOD detection performance, particularly in Near-OOD settings.

The contributions of this study are two-fold. First, we addressed the underexplored vulnerability of OOD detection under noisy label scenarios by leveraging NLL. Second, we proposed two modifications to existing NLL methods, applying model output sharpening and removing Mixup-related components, to better align them with OOD detection. These contributions provide practical insights for deploying robust OOD detection systems in real-world settings where data labeling may be noisy.

As future work, tighter integration of NLL and OOD detection could be explored to further enhance OOD detection performance. One promising direction is to design joint training objectives that simultaneously optimize both robustness to label noise and ID–OOD separability.

CRediT authorship contribution statement

Yujin Lee: Writing – original draft, Software, Methodology, Investigation, Conceptualization; **Seokho Kang:** Writing – review & editing, Validation, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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